



Dynamic defense strategies for cyber-physical systems using Stackelberg games and deep reinforcement learning in discrete and continuous time

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Abstract

As cyber threats to power grid infrastructures escalate, the urgency of understanding how to protect cyber-physical systems (CPS) has never been greater. These systems, which integrate physical processes with digital control, are increasingly susceptible to sophisticated cyberattacks that can lead to widespread disruption. While most existing defense models function within either discrete or continuous-time frameworks, this research addresses an important limitation in the literature: the limited comparative treatment of both temporal domains within a common strategic framework. This study presents a dual-domain defense framework that combines Stackelberg game theory with Deep Reinforcement Learning (DRL). Rather than merging discrete-time and continuous-time dynamics into a single hybrid model, the two temporal formulations are evaluated as alternative representations of the same CPS security interaction under identical attacker–defender scenarios and metrics, enabling a systematic comparison of strategic decision-making and physical system behavior. The objective of this framework is to facilitate proactive defense decisions that can anticipate and respond to attacks with strategic precision. We conducted extensive simulations using Python to assess the proposed model in both discrete and continuous-time scenarios. Our approach was evaluated through extensive simulations under realistic adversarial conditions to confirm its resilience and cost-effectiveness. Key findings indicate that defender-first strategies in discrete time effectively minimize system damage and alleviate computational burdens, while continuous-time responses, although immediate, demand significantly higher resource investment. This dual-domain solution offers a robust, adaptable toolset for CPS defense by clarifying the trade-offs between temporal abstractions in nonlinear and dynamic environments.

Keywords Adaptive mechanisms · Defense strategies · Dynamic modeling · Nonlinear control · Power grid security

Mathematics Subject Classification 68M10 · 91A23 · 93C10

1 Introduction

CPS are critical components of modern infrastructures, combining physical processes with digital communication networks. The growing connectivity and complexity of these systems render them increasingly vulnerable to cyber threats, especially in domains such as the power grid. Notably, False Data Injection Attacks (FDIAs) introduce malicious information into control systems, leading to incorrect decision-making processes and potential operational disruptions. These attacks may affect essential functions like voltage control or power flow within electrical networks [34]. Chen et al. [5] highlight how FDIAs target state estimation and control subsystems, such as automatic voltage control

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(AVC), by injecting data that distorts the system's state perception.

To address such threats, advanced defense mechanisms based on game theory and machine learning have been developed. Stackelberg games, in particular, offer a strategic framework for modeling adversarial interactions between attackers and defenders. In this work, we explore both directions of leadership in Stackelberg settings: one where the defender leads and the attacker follows, and the inverse, where the attacker acts first and the defender responds. This dual-role structure allows for a dynamic exploration of proactive and reactive strategies under adversarial conditions.

Simultaneously, DRL has proven to be an effective methodology for developing adaptable and intelligent defense strategies. Through environmental interaction, systems using DRL continuously learn and optimize their responses. This paper contributes by integrating DRL into the Stackelberg game structure, achieving a game-theoretic and learning-based architecture that enhances system adaptability and resilience.

A central contribution of this research is its unified modeling approach that combines both discrete-time and continuous-time representations. In contrast to earlier works such as Feng et al. [12] and Li et al. [22], which focus on only one time paradigm, our framework bridges both, providing comparative insights and increased realism in CPS modeling.

Moreover, unlike previous research where Stackelberg games were primarily applied to simplified or theoretical settings (e.g., [24, 26]), our study successfully implements these models in highly nonlinear CPS environments. This practical application demonstrates the viability of hybrid game-theoretic and learning-based solutions in real-world contexts.

An additional novel element is the detailed analysis of the dynamics of the leader-follower role. Our findings show that a defender-first strategy in discrete time leads to improved system resilience and reduced damage compared to attacker-first configurations—a distinction not addressed in prior studies such as Feng et al. [12] and Maharjan et al. [24].

The dynamic nature of the proposed system, empowered by DRL, enables real-time adaptation to evolving threats. This offers a more adaptive alternative to static defense models or less flexible strategies found in previous literature (e.g., [4, 18]).

Finally, our simulation results—conducted through extensive Python implementations—reveal clear and quantifiable benefits in terms of resilience, damage mitigation, and computational trade-offs. This practical validation provides an end-to-end evaluation of the proposed framework from earlier theoretical or limited-scope studies (e.g. [16, 41]) by providing a complete, tested defense framework suitable for real-world CPS applications.

The use of both discrete-time and continuous-time formulations in this study is intentional and addresses the dual nature of CPS security. Importantly, the two formulations are not combined into a single hybrid dynamic model; instead, they are evaluated in parallel as alternative temporal representations of the same attacker–defender interaction. The discrete-time formulation captures sequential decision epochs, policy updates, and learning-driven adaptations of the attacker and the defender, while the continuous-time formulation models the physical evolution of the CPS, enabling the analysis of stability, resilience, response time, and damage propagation. By conducting a controlled comparison under identical scenarios, leadership configurations, and evaluation metrics, the proposed framework clarifies the role and suitability of each temporal abstraction for CPS defense analysis. To clarify the objective and scope of the proposed study, the main contributions of this work can be summarized as follows:

- We develop a security-oriented modeling framework for CPS that captures adversarial interactions between attacker and defender using a Stackelberg game formulation integrated with DRL.
- We provide a systematic comparison between discrete-time and continuous-time formulations of the same CPS security interaction, evaluating them under identical attacker–defender scenarios, leadership configurations, and performance metrics.
- We analyze both defender-first (preventive commitment) and attacker-first (detection-and-response) leadership settings, clarifying their operational interpretation within CPS protection and mitigation contexts.
- We implement and validate the proposed framework through extensive simulations, assessing resilience, system damage, response time, and cost-efficiency under adversarial conditions.
- We introduce an empirically grounded convergence assessment strategy, including training stability analysis and an early stopping mechanism tailored to the multi-agent Stackelberg setting.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on CPS cybersecurity, Stackelberg games, DRL, and modeling in discrete and continuous time. Section 3 presents the proposed game-theoretic and learning-based framework, including the system dynamics and the formulation of the attacker–defender interaction under both temporal representations. Section 4 describes the simulation setup and implementation details. Section 5 reports and analyzes the experimental results, while Sect. 6 discusses implications, limitations, and inter-

pretations. Finally, Sect. 7 concludes the paper and outlines directions for future work.

2 Related work

Cybersecurity in CPS has been one of the most critical research areas of the last decade. CPS integrates physical processes with digital communication networks, creating multi-layered systems that interact and operate in real time. However, this increased connectivity and complexity also entails an increased risk of cyberattacks, directly affecting the operation of critical infrastructures such as power grids. Enhancing the security of these systems by applying advanced analysis methods, such as game theory and machine learning techniques, is now essential to address the growing threats. This section examines the current state of research in CPS, analyzing existing strategies and approaches to protect these systems from cyberattacks.

2.1 CPS cybersecurity

CPS are highly vulnerable to cyberattacks due to the complex interconnection of physical and digital infrastructures. Cyberattacks, such as FDIAs, can introduce malicious data into the control systems of power grids, misleading control mechanisms and leading to unstable or incorrect operation. Such attacks may cause severe disruptions, economic losses, and endanger user safety and infrastructure.

Recent studies have highlighted the need to develop active and passive defense measures to protect CPS [2]. The research by Ibrahim and Elhafiz [17] uses attack graph analysis to map possible intrusion paths and identify weak points that attackers can exploit [17]. Moreover, Bo et al. [2] explore active defense methods based on spatiotemporal data analysis, providing a multidimensional approach to monitoring and preventing attacks [2].

The need for comprehensive solutions has led to innovative approaches that combine the use of machine learning models and big data analytics. Studies by Hasan et al. [16] and Wang Yanpin [41] highlight the importance of integrating these technologies into control systems for real-time anomaly detection and attack prevention in smart grid CPS (SG-CPS). These approaches enable the dynamic adaptation of systems to the constantly changing conditions and threats faced by CPS.

Finally, the literature indicates that despite significant advancements in CPS cybersecurity, there are still gaps, particularly in cases where attacks exploit vulnerabilities in the communication between systems and sensors. Zhu et al. [45] analyze these gaps and emphasize the need to develop more resilient and adaptive systems capable of detecting and

preventing attacks before they reach critical infrastructure elements.

2.2 Stackelberg games in cybersecurity

Game theory, particularly Stackelberg games, has proven to be a powerful tool for analyzing the strategic interactions between defenders and attackers in CPS. In these games, the leader commits to a strategy first, and the follower subsequently optimizes its response. This hierarchical structure enables the modeling of proactive and reactive security paradigms and provides flexibility in representing adversarial interactions.

Feng et al. [12] explore the use of differential Stackelberg games for developing defense strategies in CPS power grids, focusing on attacks that could disrupt system operations. The application of differential games in continuous time enables real-time analysis of the attackers' and defenders' strategy changes, promoting more effective and flexible solutions for infrastructure protection.

Li et al. [22] and Maharjan et al. [24] examine the use of Stackelberg games for developing defense strategies in CPS and smart grids, focusing on attacks that could disrupt system operations. [22, 24] The application of Stackelberg games enables the real-time analysis of strategic interactions between defenders and attackers, promoting effective solutions for the protection of critical infrastructures. Specifically, Maharjan et al. [24] investigate demand response management in smart grids using Stackelberg games to balance energy supply and demand while addressing network security issues. Furthermore, Lye and Wing [23] explore the use of game theory to monitor the interactions between attackers and control systems, emphasizing the management of response times during attacks and maintaining the resilience of CPS. These studies highlight the effectiveness of game theory in addressing cyber threats in dynamic environments.

Beyond differential game formulations, early CPS-oriented Stackelberg implementations focused on hierarchical resilient control architectures. Early extensions of Stackelberg formulations to CPS considered hierarchical resilient control under correlated jamming attacks [44]. In particular, receding-horizon Stackelberg control laws were developed for linear discrete-time systems with constrained states and inputs, and regional exponential stability guarantees were derived under rational adversarial behavior. However, these approaches relied on quadratic program optimization and explicit system models, without incorporating adaptive learning or nonlinear value approximation mechanisms.

More recent developments extend Stackelberg formulations to wireless CPS under DDoS attacks. Peng et al. [29] formulate a dynamic stochastic zero-sum Stackelberg

game incorporating signal-to-interference-plus-noise ratio (SINR)-based communication models and Markov chain dynamics, employing refined Bayesian Nash equilibrium to address incomplete information scenarios. While this framework enhances stochastic modeling and communication-layer realism, it relies on explicit equilibrium computation and does not integrate nonlinear physical system dynamics or adaptive learning-based policy approximation.

However, the literature shows that the application of Stackelberg games in CPS still faces challenges, such as the difficulty in accurately predicting the attacker's strategy and the need for sophisticated models that incorporate learning techniques for adapting the defender's strategy. Mishra et al. [26] address these challenges by introducing a model-free reinforcement learning algorithm specifically designed for Stackelberg security games, allowing defenders to learn optimal policies without prior knowledge of attack dynamics. Similarly, Yang et al. [43] propose an adaptive learning approach to Stackelberg games, focusing on network security, which estimates unknown parameters in real time and minimizes the leader's cost under incomplete information [43]. These advancements suggest that integrating reinforcement learning with Stackelberg game models is crucial for developing dynamic, robust defense mechanisms in CPS.

In this work, the leadership roles in the Stackelberg game are interpreted operationally rather than temporally. When the defender acts as the leader, the formulation represents a *preventive* security setting, in which the defender commits in advance to a defensive policy (e.g., control hardening or mitigation intensity), anticipating that an attacker will observe these measures and adapt its strategy accordingly. Conversely, when the attacker is modeled as the leader, the framework captures a *detection-and-response* scenario, where the defender reacts after observing the attack's impact. This distinction clarifies that leadership does not imply inherent dominance, but reflects different security paradigms.

2.3 DRL in CPS

DRL has emerged as a powerful tool for developing defense strategies in CPS, particularly in power grids. The application of DRL algorithms, such as the actor-critic algorithm, allows systems to dynamically learn optimal response strategies to cyberattacks through interaction with the environment and continuous parameter updates.

Studies by Chatterjee and Namin [4] examine the use of DRL to detect malicious activities on websites, highlighting the real-time effectiveness of the algorithm. Wang et al. [41] propose applying DRL in network control systems to optimize network operations and protect against attacks.

Additionally, studies by Hasan et al. [16] and Su et al. [36] demonstrate how DRL can be applied to nonlinear dynamic systems for developing optimal defense policies. Through

the adaptability of the algorithm, systems can dynamically adjust their strategies, leveraging learning techniques and big data analytics to improve defense policy effectiveness.

However, there are challenges in implementing DRL, such as the need for large volumes of data and the requirement for real-time response, making its application difficult in complex and multidimensional CPS. Studies by Ibrahim and Elhafiz [18] emphasize that integrating DRL into CPS requires advanced techniques to optimize model training speed, while Bo et al. [2] explore approaches to improve network resilience through DRL techniques.

Despite the challenges, contemporary research shows that combining DRL with game theory can provide new and innovative solutions to addressing attacks in CPS, promoting a multidimensional approach that integrates prediction and real-time response techniques.

2.4 Approaches in discrete and continuous time

The analysis of cybersecurity strategies in CPS requires the application of models in both discrete and continuous time, depending on the system's nature and application needs. Discrete time approaches are useful for analyzing strategies based on predefined time steps, allowing the simulation and analysis of the system under more controlled conditions. In contrast, continuous time approaches are ideal for simulating the dynamic evolution of the system in real-time, providing a more accurate and realistic analysis of attacker-defender interactions.

Feng et al. [12] and Li et al. [22] have analyzed the use of differential Stackelberg games for modeling and analyzing defense strategies in CPS power grids in continuous time. These approaches allow for the analysis and optimization of strategies in real-time, which is crucial for effectively responding to attacks on systems that require immediate reactions. Their study shows that by applying these models, defense strategies can adapt promptly to changes caused by the attacker's actions, enhancing overall system security.

Maharjan et al. [24] examine the applications of these approaches in discrete time, analyzing how strategies can be precisely formulated and controlled at each time step. The use of discrete models provides a more stable and predictable basis for developing strategies, allowing the integration of specialized policies based on data from previous time steps.

While discrete approaches offer incremental analysis, CPS also need to adapt to continuous time conditions to respond to attacks in real-time. Lee et al. [21] emphasize that combining discrete and continuous models in CPS defense strategies, using System of Systems (SoS) approaches and hybrid modeling techniques, enhances adaptability and resilience. Their use of DEVS BUS for synchronizing these models allows CPS to dynamically respond to immediate threats effectively.

Research by Hasan et al. [16] emphasizes that optimizing the application of strategies in continuous time is critical for achieving high levels of security in CPS power grid and Ibrahim [18] also analyze these strategies, focusing on system resilience and real-time adaptation.

2.5 Simulation with python

Python has become one of the primary tools for simulating and developing cybersecurity strategies in CPS. With its advanced libraries, such as TensorFlow and PyTorch, it provides a powerful environment for developing machine learning models and simulating strategies in real time. Python's flexibility allows for dynamic model customization, offering a platform for analyzing complex systems and evaluating game-theoretic strategies.

Studies by Ibrahim [18] and Wang [41] have integrated the use of Python to analyze CPS power grid security, focusing their research on simulations that employ machine learning techniques for attack detection and prediction. The results showed that Python-based simulations offer accuracy and real-time adaptability, making it a reliable platform for developing defense strategies.

Hasan et al. [16] also used Python to develop simulations combining Stackelberg games and DRL techniques, while Li et al. [22] developed a dynamic simulation platform focused on analyzing FDIA attacks. This platform allowed for the study of attacker-defender interactions in a controlled environment, providing a basis for developing adaptive defense strategies in CPS.

Moreover, studies by Chatterjee and Namin [4] used Python for malicious website detection, highlighting Python's flexibility in developing detection systems using machine learning algorithms. The ability to integrate different models and dynamically configure strategies enables systems to adapt to current threat conditions, improving the resilience of CPS.

The literature indicates that, despite Python's capabilities, there are challenges related to the performance and scalability of simulations in large-scale CPS. Studies by Bo et al. [2] emphasize the need to optimize algorithms to increase execution speed and reduce model complexity, ensuring Python's effective use as a tool for developing CPS simulations.

2.6 Applications of machine learning and challenges

Integrating machine learning techniques into CPS is a crucial step for real-time attack detection and prevention. These techniques enable continuous data monitoring and analysis for identifying anomalies and malicious actions, thus offering a dynamic and adaptive defense. Studies by Hasan et al. [16] and Bo et al. [2] demonstrate how machine learning can

be integrated into CPS to develop adaptive defense strategies that respond to changing conditions.

However, implementing these techniques is not without challenges. One of the key issues is the need for large, real-time data sets required for training machine learning algorithms. The lack of such data and the complexity of systems make accurate attack detection challenging, as noted by Ibrahim and Elhafiz [18]. Furthermore, studies by [1] and [35] highlight that the scalability of these algorithms is a challenge, especially when applied to large-scale CPS, such as power grids.

To address these issues, Wang et al. [41] propose hybrid approaches that integrate Stackelberg games with DRL algorithms. This integration enables CPS to dynamically adapt their defense strategies by leveraging real-time data and feedback, thereby enhancing system resilience. Similarly, Li et al. [22] develop machine learning-based simulation frameworks for attack prediction and strategy optimization, demonstrating that dynamic adaptation can yield significant performance and security benefits.

In parallel, the research by Lye and Wing [23] and the recent work by Chatterjee and Namin [4] examine the use of machine learning algorithms for threat detection in industrial control systems, emphasizing the need for developing algorithms that are resilient to evolving threats. This flexibility is critical for ensuring the long-term protection of CPS.

While existing approaches have demonstrated their effectiveness, optimizing algorithms for speed and integrating dynamic learning techniques remain priorities for improving CPS performance, as noted by Bo et al. [2]

2.7 Nonlinear cyber dynamics systems

Nonlinear dynamic systems present a particularly demanding area in the analysis and security of CPS. These systems, characterized by complex and unpredictable behavior, are vulnerable to cyberattacks that can exploit the nonlinear interactions among their subsystems. Such attacks are often difficult to predict and prevent, necessitating the development of specialized defense strategies.

Studies by Wang et al. [41] and Li et al. [22] explore the use of machine learning algorithms, such as DRL, for analyzing and developing defense strategies in nonlinear systems. Through these algorithms, CPS can detect changes in system behavior and dynamically adjust their strategies to counter cyberattacks. These studies focus on enhancing system adaptability through the analysis of nonlinear behavior.

Additionally, the literature indicates that Stackelberg game theory can be applied to nonlinear dynamic systems to analyze the interaction between attackers and defenders. Hasan et al. [16] propose integrating approaches that combine DRL and Stackelberg games to develop adaptable and flexible strategies in nonlinear environments. Furthermore,

studies by Lye and Wing [23] examine how combined strategies can incorporate nonlinear characteristics for better attack prediction and prevention.

While several studies apply Stackelberg formulations to CPS security, many focus on communication-layer attack models. For example, Wang et al. [42] derive equilibrium strategies for multi-channel DoS attacks using energy-constrained optimization. Although analytically rigorous, such approaches abstract from nonlinear physical system dynamics, which remain critical for assessing stability and resilience in power-grid CPS.

Despite these advancements, Ibrahim and Elhafiz [18] emphasize that formulating effective strategies for nonlinear systems remains a challenge due to their uncertainty and unpredictability. Bo et al. [2] also highlight the need for more flexible and faster learning methods that respond to the complexities of these systems.

This research aims to bridge these gaps by integrating nonlinear analysis models with DRL techniques and game theory to develop adaptable and dynamic defense strategies, offering a more realistic approach to enhancing CPS security.

2.8 Resilience and recovery in CPS

The resilience and recovery of CPS after cyberattacks are crucial elements for protecting infrastructures and ensuring their uninterrupted operation. Despite efforts to prevent attacks, it is essential for CPS to be able to adapt, minimize the impact of attacks, and recover their functionality quickly and effectively.

In their paper, Lye and Wing [23] analyze game-theoretic strategies for network security, focusing on the interactions between an attacker and a network administrator. They model these interactions as a two-player stochastic game, where both parties aim to optimize their respective strategies. The attacker attempts to exploit the network, while the administrator tries to defend it. The paper uses a nonlinear program to compute Nash equilibria, which represent the best-response strategies for both the attacker and the administrator. By applying this model, they provide insights into realistic strategies that administrators can use to enhance network security.

Zhu et al. [45] and Malik et al. [25] emphasize the importance of integrating backup strategies and proactive measures to enhance CPS resilience. These strategies include the proactive analysis of system vulnerabilities and the integration of backup mechanisms that activate when an attack is detected.

Additionally, Hasan et al. [16] examine how machine learning techniques, such as DRL, can be integrated into recovery systems for automatically adapting strategies in real-time and quickly restoring system functionality. This dynamic approach allows CPS to detect and respond immediately to changes caused by attacks.

The literature also shows that strengthening CPS resilience involves developing strategies for resource redirection and system reconfiguration. Studies by Wang et al. [41] and Chatterjee and Namin [4] suggest using artificial intelligence to adapt strategies and manage resources, ensuring uninterrupted service provision even in cases of successful attacks.

This research extends the study of CPS resilience by integrating innovative methods that combine Stackelberg games and DRL techniques to create dynamic recovery strategies. This approach ensures that CPS can adjust their strategies and restore functionality effectively, thus contributing to the enhancement of critical infrastructure resilience.

2.9 Gaps in the literature and research contribution

Although analytical Stackelberg formulations such as those in [29, 42, 44] provide stability guarantees and equilibrium characterizations under jamming or DoS attacks, they rely on explicit optimization and equilibrium derivation. The integration of nonlinear physical dynamics with adaptive learning-based equilibrium approximation remains comparatively underexplored.

Despite significant progress in the research of CPS and cybersecurity, there are still gaps that require further investigation. Many existing studies focus exclusively on either discrete or continuous time, while few combine the two approaches to provide a holistic analysis of attacks and defense strategies in CPS.

The research by Li et al. [22] highlights the lack of integration of Stackelberg games with machine learning techniques in continuous time, which is critical for addressing cyberattacks in dynamic environments. Moreover, the use of DRL remains limited, with existing approaches primarily focusing on industrial control systems rather than large-scale CPS power grids.

Ibrahim and Elhafiz [18] also emphasize the need for more flexible and dynamic strategies that can adapt quickly to evolving threats, as current models do not provide sufficient adaptability to the continuously changing environment. This research indicates that existing systems require significant improvements to increase response speed and the effectiveness of defense policies.

This work contributes to addressing these gaps a framework that integrates both Stackelberg games and DRL techniques for analyzing and developing defense strategies in CPS power grids. Our research combines approaches in discrete and continuous time, integrating data from multiple sources to achieve a more realistic and flexible analysis. Additionally, through Python simulations, our study validates the effectiveness of the strategies in real-time, offering an innovative approach to enhancing CPS security.

The analysis of CPS and cybersecurity reveals the complexity and challenges of managing and protecting these

systems. The literature highlights the importance of integrating approaches that combine game theory, machine learning techniques, and dynamic analysis in both discrete and continuous time. The use of DRL and Stackelberg games, as well as their integration into nonlinear dynamic systems, offers innovative solutions for enhancing the security and resilience of CPS.

Despite advancements, there are still gaps that require further research, particularly concerning resilience management and system recovery after cyberattacks. This work aims to bridge these gaps by applying innovative methods, proposing a hybrid approach that integrates different techniques and strategies to address attacks and ensure the functionality of CPS in real-time. In this way, we aim to create a more comprehensive framework for studying and enhancing the security of critical infrastructures.

3 Stackelberg game modeling framework

This study presents a dynamic defense strategy for CPS, with a focus on power grids. We address both discrete and continuous time scenarios to capture the varying dynamics of real-world systems. The interaction between the defender and the attacker is modeled using a Stackelberg game framework, allowing for alternating leadership roles.

3.1 The structure of the CPS

We live in the era of the fourth industrial revolution which means many technological and digital advantages and disadvantages. This incredible phenomenon has changed not only our daily lives but also the way that enormous enterprises and systems operate. CPS are increasingly being integrated into power grids to enhance their efficiency, reliability, and sustainability.

To avoid ambiguity, we emphasize that the discrete-time and continuous-time models are treated as alternative temporal representations of the same CPS security interaction. Both formulations are evaluated under identical leadership configurations (defender-first and attacker-first) and are assessed using the same set of system-level and learning-related metrics, including resilience, system damage, response time, critic losses, and cost efficiency. This design ensures a fair and consistent benchmarking procedure, where observed performance differences can be attributed to the temporal modeling choice rather than to changes in scenarios or objectives.

The CPS environment comprises critical components of a power grid, including generators, transmission lines, control systems, and communication networks. The system is vulnerable to cyberattacks that can disrupt physical operations, leading to potential blackouts or equipment damage.

These systems are known as CPS, and there are many other synonymous phrases that characterize these systems, the most notable of which are as follows:

One such term is the Integrated Energy System (IES), which refers to systems that combine diverse energy sources and technologies to optimize energy usage and distribution. These systems often incorporate renewable energy sources, energy storage solutions, and smart grid technologies to manage energy from various sources efficiently, optimize energy use in buildings and communities, and develop smart infrastructure for improved energy distribution and consumption [13].

The term Smart Grid System (SGS) is also prevalent in this context. SGS integrates modern communication and information technologies with traditional power grids to improve the efficiency, reliability, and sustainability of electricity services. Applications of SGS include managing and optimizing electric power grids, integrating renewable energy sources, and developing smart metering and energy consumption monitoring systems [10].

A CPS is a system that seamlessly combines computational algorithms and physical components, interconnected through a network of sensors and actuators. The correct operation of a CPS is highly dependent on the precise timing and synchronization between its digital and physical elements. Time plays a vital role in ensuring that these systems respond promptly and accurately to changes in their environment [20].

The basic layers of CPS are: the physical world, cyber world, and human contribution [45]. The layers and subcategories are presented in great detail in Fig. 1. The physical layer and the control layer are parts of the physical world. The physical layer consists of the buildings and the facilities known as Critical Infrastructures. The Council of the European Union in its directive [38] has adopted the same term. This directive identifies and designates critical infrastructures in Europe and the assessment of the need to improve their protection. The control layer plays a critical role in a CPS by integrating sensing, actuation, and security monitoring functionalities, including components such as sensors, actuators, and an Intrusion Detection System (IDS). This layer of interfaces is the bridge between the physical and cyber worlds [6]. Sensors receive data from the physical layer and transmit it to the cyber world. The actuators are the 'digital eyes' of the systems, they activate alarms. They receive processed data from the cyber world. The components of cyber world are communication and network layers that provide real time data cloud storage, analysis, etc. The last level of structure of the CPS is the Human layer. This component is divided into two categories: the management and the supervisory layer. The essential decisions are made by humans in this layer.

As illustrated in Fig. 1, the *Counteraction Layer* constitutes the operational component of the proposed CPS security

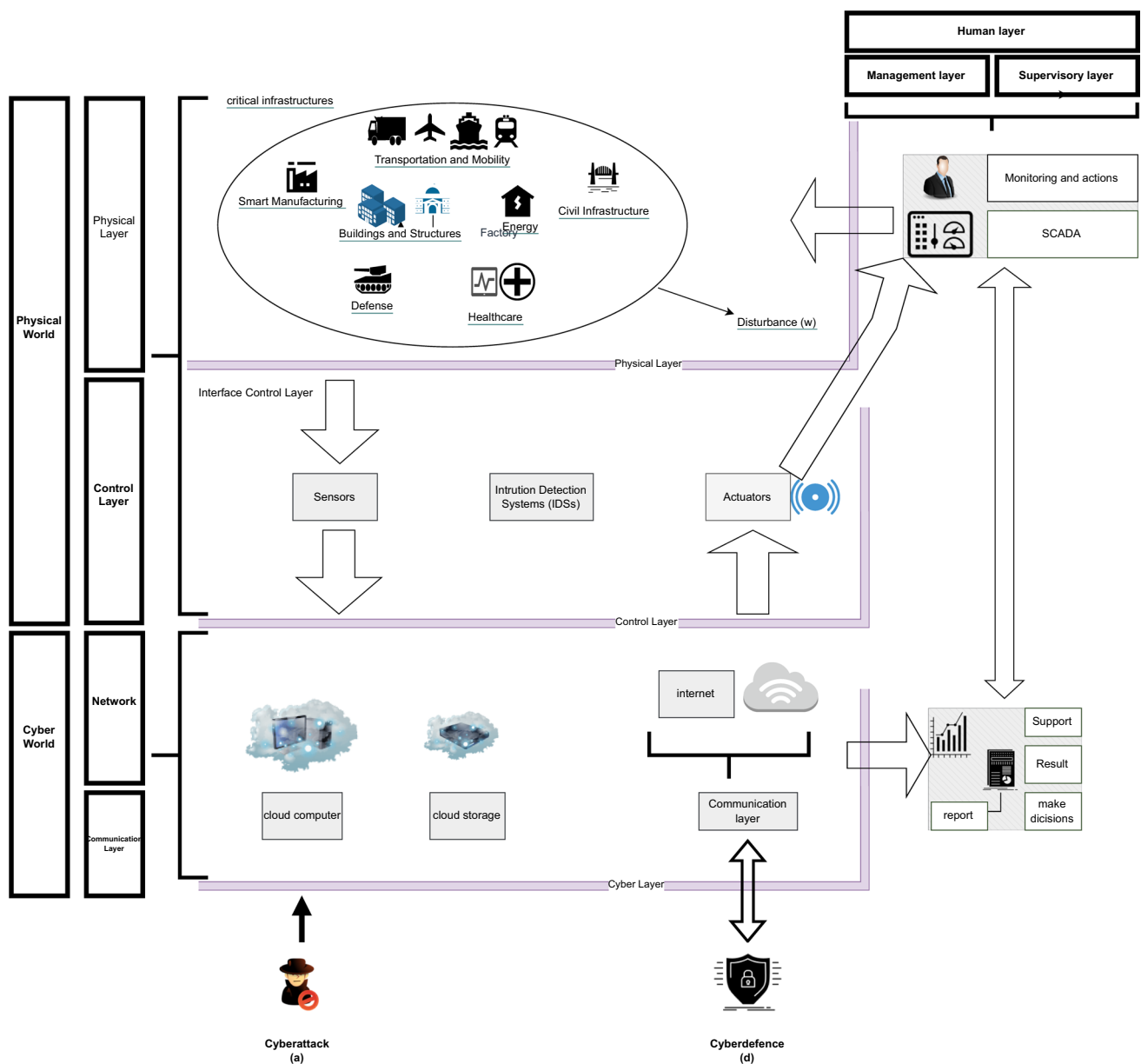


Fig. 1 High-level architecture of the proposed CPS security framework. The Counteraction Layer implements defensive mitigation actions derived from the Stackelberg game formulation and the DRL-based controller. Under the sequential Stackelberg information structure, the

Leader selects its action from the observed system state, whereas the Follower conditions its response on both the current state and the realized Leader action

framework. Its role is to translate the defender's selected strategy—derived from the Stackelberg game formulation and the DRL-based controller—into concrete mitigation actions applied to the CPS. These actions may include protective control adjustments, containment measures, or recovery-oriented responses, aiming to counteract the attacker's impact on the physical process and to support system stability, resilience, and damage mitigation.

3.2 Stackelberg game framework

To model the strategic interactions between the defender and the attacker, we employ a Stackelberg game with two scenarios based on leadership roles:

In a two-player Stackelberg game, a hierarchical relationship exists between the players, where one player, known as the leader, makes decisions first, and the other player, the follower subsequently responds using the current system state and the realized action selected by the leader. This structure

holds in scenarios where there is a clear distinction between the leader and follower roles and where sequential decision-making is feasible and meaningful. In a Stackelberg game, equilibrium occurs under specific conditions:

1. **Leadership advantage:** The leading player (Stackelberg leader) must possess a strategic advantage over the follower. This advantage enables the leader to set their strategy first, knowing the follower's subsequent response.
2. **Optimal responses:** Both players must make optimal decisions given the actions of the other player. The leader's strategy should maximize their utility considering the follower's response, and similarly, the follower's strategy should maximize their utility given the leader's chosen strategy.
3. **Consistency:** The strategies chosen by both players must be consistent with the game's rules and constraints. This ensures that the equilibrium reached is feasible and stable.

These conditions collectively ensure that the equilibrium achieved in a two-player Stackelberg game is both rational and stable. Building upon the foundations laid by [11], the adversarial interaction within CPS is modeled using differential game concepts. While several prior works formulate CPS security as a zero-sum differential game under Nash equilibrium principles, the present study adopts a hierarchical general-sum Stackelberg formulation. This paper aims to advance the understanding by presenting a two-player game theoretical model for CPS, delineating the dynamics between defender and attacker.

Although previous studies [11, 19, 39, 40] have predominantly examined CPS through the lens of Nash equilibrium, limited attention has been devoted to the exploration of the Stackelberg equilibrium. Accordingly, this paper seeks to introduce the Stackelberg theory within the context of CPS defense, where the defender assumes the role of a leader and the attacker that of a follower.

To model the strategic interactions between the defender and the attacker, we employ a Stackelberg game with two scenarios based on leadership roles:

Scenario 1: Attacker as leader

Attacker's Move: The attacker selects an optimal strategy to maximize disruption, targeting specific components or communication channels within the CPS. **Defender's Response:** After the attacker selects its action, the defender computes its response using the current system state together with the realized attacker action, aiming to minimize system damage and restore normal operations.

Scenario 2: Defender as leader

Defender's Move: The defender proactively deploys defense mechanisms, anticipating potential attack strategies.

Attacker's Response: After the defender selects its preventive action, the attacker computes its response using the current system state together with the realized defender action, aiming to maximize its impact despite the deployed protective measures.

Both scenarios are analyzed in discrete and continuous time settings to understand the temporal effects of strategy implementation.

The proposed framework adopts a sequential Stackelberg information structure. At each decision step, the Leader selects its action using the current system state, $u_t^L = \pi_L(x_t)$. The Follower then computes its response using both the current state and the realized Leader action, $u_t^F = \pi_F(x_t, u_t^L)$. Thus, the Follower observes the Leader's current realized action, but does not observe the Leader's policy parameters, neural-network weights, or internal decision mechanism. This conditional information structure is applied consistently in both the attacker-leader and defender-leader configurations and corresponds to the sequential implementation used in the algorithm.

3.3 The model

3.3.1 General CPS state-space formulation

Based on the relevant literature [8, 11, 27, 33, 45] we develop a model that replaces the use of the Nash equilibrium with a Stackelberg equilibrium, focusing on the role of time t . Following the article [9], the Nash equilibrium implies the Stackelberg equilibrium. We consider two players; the first is the attacker (α) and the second is the defender (d).

Following the formulation proposed by Zhu et al. [45] and building upon Feng et al. [11] and Vamvoudakis et al. [40], we denote the state of the system at time t by $x(t) \in \mathbb{R}^n$, and define the primary state dynamics of the CPS as follows:

$$\dot{x}(t) = f(x(t)) + g(x(t))\alpha(t) + k(x(t))d(t), \quad (1)$$

with initial condition $x(t_0) = x_0$. In the above, f represents the dynamics of the system in the absence of any external influence, α corresponds to the attackers' actions with g modeling the response of the system to these actions, and d corresponds to the defenders' actions with k modeling the response of the system to these actions. The dynamical system (1) will be used as the formulation in the subsequent analysis.

The dynamical system (1) provides a general formulation encompassing a wide range of instances of CPS and the possible interactions between various agents (e.g. attackers or defenders etc).

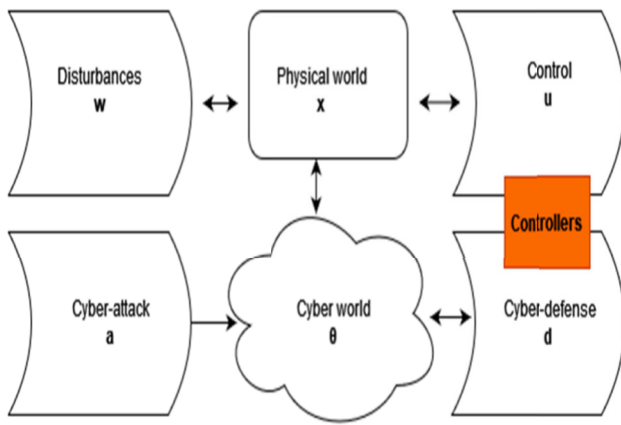


Fig. 2 The dynamics of cyber attack and Cyber defense in a CPS

3.3.2 Continuous-time power-grid dynamics

The dynamics of the CPS in continuous time, particularly for a power grid, are modeled using Equation (1), which captures both the natural behavior of the system and the effects of control actions of the attacker and defender. In this context, the state vector is defined as

$$\mathbf{x}(t) = \begin{bmatrix} \delta(t) \\ \omega(t) \end{bmatrix},$$

where $\delta(t)$ represents the angle of the rotor and $\omega(t)$ represents the frequency deviation. These variables are critical in describing the state of a power grid, as they determine the synchronization and stability of the system. The natural dynamics of the system are described by the function.

$$\mathbf{f}(\mathbf{x}(t)) = \begin{bmatrix} \omega(t) \\ \frac{P_m - P_{\max} \sin(\delta(t)) - D \cdot \omega(t)}{M} \end{bmatrix},$$

which characterizes the generator's behavior in the power grid, where P_m is the mechanical power, $P_{\max}$ is the maximum electrical power output, D is the damping coefficient and M is the inertia constant. These physical properties define how the grid responds to disturbances and external influences.

The effects of the attacker and the defender control actions are incorporated through the functions.

$$\mathbf{g}(\mathbf{x}(t)) = \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} \quad \text{and} \quad \mathbf{k}(\mathbf{x}(t)) = \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix},$$

which model how the attacker's and defender's interventions impact the frequency deviation of the system. The control actions, $a(t)$ and $d(t)$, represent the respective strategies of the attacker and defender at time t . These functions capture the dynamics of the power grid under both normal operation and cyber-attack conditions, enabling the study of system

resilience and stability under different attack-defense scenarios.

The dependence of each control input on the available information is determined by the leadership assignment. If the defender is the Leader, its control is represented as $d(t) = \pi_d^L(x(t))$. If the defender is the Follower, its response is represented as $d(t) = \pi_d^F(x(t), a(t))$, where $a(t)$ is the realized action selected by the attacker acting as Leader. Analogously, when the attacker is the Follower, its policy is conditioned on both the current state and the realized defender action. The same sequential information structure is used in the discrete-time implementation.

3.3.3 Discrete-time CPS dynamics

The discrete-time dynamics of the system is described by the following equation:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t [\mathbf{f}(\mathbf{x}_k) + \mathbf{g}(\mathbf{x}_k)a_k + \mathbf{k}(\mathbf{x}_k)d_k], \quad (2)$$

where:

- $\mathbf{x}_k$ is the state vector at time step k .
- a_k is the action of the attacker at time step k .
- d_k is the action of the defender at time step k .
- Δt is the time step size.
- $\mathbf{f}(\mathbf{x}_k)$ represents the natural dynamics of the system without control inputs.
- $\mathbf{g}(\mathbf{x}_k)$ captures the influence of the attacker's action on the system.
- $\mathbf{k}(\mathbf{x}_k)$ captures the influence of the defender's action on the system.

For our specific system, the state vector and functions are defined as follows:

State vector:

$$\mathbf{x}_k = \begin{bmatrix} \delta_k \\ \omega_k \end{bmatrix}$$

Natural dynamics $\mathbf{f}(\mathbf{x}_k)$:

$$\mathbf{f}(\mathbf{x}_k) = \begin{bmatrix} \omega_k \\ \frac{1}{M} (P_m - P_{\max} \sin(\delta_k) - D\omega_k) \end{bmatrix}$$

Attacker influence $\mathbf{g}(\mathbf{x}_k)$:

$$\mathbf{g}(\mathbf{x}_k) = \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix}$$

Defender influence $\mathbf{k}(\mathbf{x}_k)$:

$$\mathbf{k}(\mathbf{x}_k) = \begin{bmatrix} 0 \\ \omega_k \\ -\frac{\omega_k}{M} \end{bmatrix}$$

Therefore, the discrete-time update equations become:

$$\begin{aligned} \delta_{k+1} &= \delta_k + \Delta t \cdot \omega_k, \\ \omega_{k+1} &= \omega_k + \Delta t \left[\frac{1}{M} (P_m - P_{\max} \sin(\delta_k) - D\omega_k) \right. \\ &\quad \left. + \frac{1}{M} a_k - \frac{\omega_k}{M} d_k \right]. \end{aligned}$$

where:

- δ_k is the rotor angle at time step k .
- ω_k is the rotor speed at time step k .
- M is the inertia constant.
- D is the damping coefficient.
- P_m is the mechanical power input.
- $P_{\max}$ is the maximum electrical power output.
- a_k is the attacker's action aimed at destabilizing the system.
- d_k is the defender's action aimed at stabilizing the system.

This formulation captures the dynamics of the system under the influence of both the attacker and the defender, as implemented in the code. It provides a clear and general representation suitable for inclusion in the paper.

3.3.4 Interpretation of the defender input across temporal formulations

The apparent difference in the dependence of the frequency dynamics on the defender action between the continuous-time and discrete-time formulations is due to the temporal representation and implementation of the control input. In continuous time, the defender action $d(t)$ enters the dynamics as a direct control input, reflecting an instantaneous mitigation effort applied to the system.

In contrast, in the discrete-time formulation, the defender action d_k is implemented as a state-dependent feedback term that effectively modulates the damping of the system during each time step. This representation arises naturally from the time discretization and is commonly adopted to enhance numerical stability and learning convergence in reinforcement learning-based implementations. Despite this difference in mathematical form, both representations capture the same physical role of the defender in counteracting frequency deviations and preserving system stability.

3.3.5 Operational cybersecurity interpretation

An example of this interaction between the Energy Grid Operator and Cyber Adversaries within a real world CPS (for an energy grid) is provided in Fig. 2. The Energy Grid Operator is responsible for the secure and stable operation of the energy grid, acting as the primary controller and defender of the system. The controllers and cyber-defense (d) elements represent this role in the Figure. The control (u) function reflects the actions taken by the operator to regulate and manage the physical system (e.g., power generation and distribution), while cyber-defense (d) encompasses the security measures implemented to detect, prevent, and mitigate cyber attacks. In practice, the Energy Grid Operator constantly monitors the system, identifies potential threats, and responds to any attempts to compromise the system's integrity.

Cyber-adversaries represent the entities that aim to exploit vulnerabilities within the CPS to disrupt or compromise the grid. This role is illustrated by the cyber-attack (α) pathway, which attacks directly into cyber world (θ), as represented in Fig. 2. These adversaries may be nation-state actors, hacktivists, insider threats or financially motivated cybercriminals. They target the grid to disrupt operations by compromising data, interrupting service or seizing unauthorized control of key infrastructure. Cyber adversaries often leverage various techniques such as malware, phishing and DDoS to bypass the system defences creating disruptions. The diagram shows the workings of both the energy grid operator and the cyber adversaries. While cyberattacks (α) target the system's cyber world (θ), attempting to impact the physical world (x) through disruption, the Energy Grid Operator utilizes Controllers and Cyber-defense (d) to counter these attacks and preserve grid stability. This emphasizes the ongoing tug of war between offense and defense, which highlights the importance of sound cybersecurity mechanisms in even prestigious CPS offerings. In other words, Fig. 2 shows a more complex network of interactions necessary to depict the roles and actions that take place in practice between the energy grid operator (left side) and cyber adversaries (right).

3.3.6 Stackelberg game interpretation

We model the interaction between the defender and attacker using the Stackelberg game. The Stackelberg equilibrium is the best suited type of game equilibrium type for the type of situation we wish to model. In Stackelberg equilibria, the two agents (defender and attacker in our case) are not treated as equal, with one agent typically acting first and having a dominant role in the evolution of the game (called the leader) and the other (called the follower) responding to the action of the leader. The difference in their role, may put one (the leader) in superior position to the other (the follower), and may also reflect difference in the information

concerning the game between them. In security, it is natural to assume that the attacker (a) will act first, having full control over the decision of when to act, in contrast to the defender (d) that typically acts in response to the action of the attacker and with the aim of minimizing the adverse effects of the attackers action. Hence, typically in security games, the attacker (a) acts as a leader while the defender (d) acts as a follower, rendering the protocol of the Stackelberg game the suitable game theoretic equilibrium notion. Within this framework, it is acceptable that the role of the leader and follower may alternate when the game is played repeatedly, as for example in the case of a major attack, after which there may be a major upgrade of the defense measures of the system, putting the defenders into the role of the leader, with the attacker having to reformulate its strategies to cope with passing through the new augmented security framework of the system. The repeated game can either be performed in continuous or discrete time. The qualitative information provided by both formulations (continuous or discrete time) is similar, while each of them has its merits. For example, the continuous formulation, may lead to solutions for the optimal policy in closed form, building upon the well established tools from differential equations and control theory. On the other hand, the discrete time formulation, is more realistic in the sense that the nature of the actions of both agents takes place in discrete time instants. Moreover, the numerical implementation of the game, including the DRL formulation, is naturally treated in the discrete time formulation, often however, as a discretization of the solutions and strategies obtained from the continuous time formulation. For the above reasons, and for the completeness of the present paper, we decide to (briefly) address both.

Defender and attacker payoff functions

The continuous-time payoff functional for each agent $i \in \{\alpha, d\}$ is defined as

$$J_i = \int_0^\infty e^{-\rho t} G_i(x(t), u_i(t)) dt,$$

where $\rho > 0$ is the continuous-time discount rate. In the numerical implementation, the corresponding discrete discount factor satisfies $\gamma = e^{-\rho dt}$.

The stage-wise payoff functions are defined as

$$\begin{aligned} G_d(x(t), d(t)) &= -\omega(t)^2 + \lambda|d(t)|, \\ G_a(x(t), a(t)) &= \omega(t)^2, \end{aligned}$$

where $\lambda > 0$ is a weighting parameter that penalizes excessive defensive effort.

Remark 1 We emphasize that $\omega(\cdot)$ and $\delta(\cdot)$ are solutions of (1) (recall that $x = (\delta, \omega)^T$), and therefore depend implicitly

on the chosen control actions $a(\cdot)$ and $d(\cdot)$. Strictly speaking, the state trajectory should be written as $x(t) = x(t; a, d)$, i.e. $\omega(t) = \omega(t; a, d)$ and $\delta(t) = \delta(t; a, d)$. However, in order to avoid excessive notational complexity, we adopt the simplified notation $x(t)$ throughout the paper, while keeping this functional dependence in mind.

The defender therefore aims to minimize frequency deviations while balancing control effort, whereas the attacker aims to maximize frequency oscillations.

Although the system dynamics in (1) are expressed in a control-theoretic state-space form, they are used to model cyber attacks and defenses at an abstract but operational level. Cyber attacks such as denial-of-service (DoS), data manipulation, or communication disruptions are not represented explicitly at the packet or network layer. Instead, their impact is captured through the attacker input $\alpha(t)$, which models the effective disturbance induced on the CPS and its physical process.

In particular, $\alpha(t)$ represents the aggregated effect of cyber adversarial actions that degrade sensing, actuation, or control availability, ultimately perturbing the system state evolution. Similarly, the defender input $d(t)$ does not represent intrusion detection, but rather mitigation and response mechanisms, such as control reconfiguration, damping adjustments, load shedding, or other protective actions applied to counteract the observed impact of the attack. Defensive actions are selected according to the relevant leadership configuration. When the defender acts as Leader, its action depends on the observed system state. When the defender acts as Follower, its response depends on both the current system state and the realized attacker action. This formulation represents the sequential information flow adopted in the implemented Stackelberg architecture.

This abstraction enables the analysis of attack–defense interactions at the system-dynamics level, where stability, resilience, and performance degradation are the primary security concerns.

Discrete-time payoff functions

For numerical implementation, the continuous-time dynamical system describing the evolution of the state is discretized as in (2), the cost functionals are discretized in time (see below), and for the application of the DRL framework the associated value functions are represented implicitly via the discounted Bellman recursion. We also emphasize that throughout this section the state of the system is implicitly assumed to be a function of the chosen controls a and d (or the policy functions π_a and π_d) even though for the sake of decluttering the notation we do not indicate this explicitly (see Remark 1).

The payoff function for the defender, J_d , is chosen so as to minimize oscillations while encouraging active defensive responses:

$$J_d = \sum_{k=0}^N r_d(k) = -\sum_{k=0}^N \omega_{\text{next}}^2 + 0.1 \sum_{k=0}^N |u_d(k)|$$

where $r_d(k) = -\omega_{\text{next}}^2 + 0.1|u_d(k)|$. The first term penalizes oscillations (captured by ω_{next}^2), while the second term rewards the defender for proactive actions. This payoff structure helps the defender to stabilize the system.

For the attacker, the payoff function J_a is defined to maximize oscillations:

$$J_a = \sum_{k=0}^N r_a(k) = \sum_{k=0}^N \omega_{\text{next}}^2$$

where $r_a(k) = \omega_{\text{next}}^2$. This formulation incentivizes the attacker to increase system instability, maximizing oscillatory behavior to disrupt the defender's control.

We also keep in mind the above Remark 1, and emphasize that ω depends on the chosen actions a and d .

The cumulative payoff functions J_d and J_a allow each agent to achieve their objectives (stability for the defender, disruption for the attacker), while the system's dynamics evolve in response to their actions at each discrete time step.

Stackelberg equilibrium

In the Stackelberg game framework, one of the two agents, either the attacker or the defender takes the leadership role (henceforth denoted by L), setting their strategy first, leaving the role of the follower (henceforth denoted by F) to the other. The leader L chooses his/her action first anticipating the response of the follower F .

When the defender is the leader, the strategy $d(t)$ is first optimized to stabilize the system, anticipating the response $a(t)$ from the attacker. Conversely, when the attacker is the leader, $a(t)$ is optimized first, aiming to destabilize the frequency, with the defender subsequently choosing $d(t)$ in response to minimize deviations in $\delta(t)$ and $\omega(t)$.

In either case, the resulting state trajectory $x(t)$ evolves according the state dynamics Eqs. (1) or (2) (depending on whether we consider the continuous or the discrete time case), and implicitly depends on the chosen control strategies, as clarified in Remark 1. This hierarchical optimization leads to a Stackelberg equilibrium. The very nature of CPS security games, makes the Stackelberg notion of equilibrium, obtained through the afore mentioned hierarchical optimization problem, the proper notion of equilibrium for the application in consideration.

We now proceed to the mathematical Formulation of the Stackelberg equilibrium (SE) and define the corresponding game value.

Let $\mathcal{A}$ and $\mathcal{D}$ denote the admissible strategy spaces of the attacker and defender, respectively. Strategies are assumed to follow the sequential information structure $u^L(t) = \pi_L(x(t))$ and $u^F(t) = \pi_F(x(t), u^L(t))$. Accordingly, in the attacker-leader case, $a(t) = \pi_\alpha^L(x(t))$ and $d(t) = \pi_d^F(x(t), a(t))$, whereas in the defender-leader case, $d(t) = \pi_d^L(x(t))$ and $a(t) = \pi_\alpha^F(x(t), d(t))$.

As already mentioned the role of the leader (L) and the follower (F) may interchange, with either the attacker or the defender assuming each role. For clarity we analyse both cases below:

Attacker-leader case For any admissible attacker strategy $a(\cdot) \in \mathcal{A}$, the defender (follower) solves the lower-level optimization problem

$$d^*(a) = \arg \max_{d(\cdot) \in \mathcal{D}} J_d(a, d).$$

Anticipating this best-response mapping $d^*(a)$, the attacker (leader) solves the upper-level problem

$$a^* = \arg \max_{a(\cdot) \in \mathcal{A}} J_\alpha(a, d^*(a)).$$

The Stackelberg equilibrium is therefore defined as the hierarchical pair

$$SE := (a^*, d^*(a^*)).$$

The corresponding value function of the Stackelberg game is defined as

$$V^\alpha(x_0) = \sup_{a(\cdot) \in \mathcal{A}} J_\alpha(a, d^*(a)),$$

which represents the maximum cumulative payoff achievable by the attacker under optimal follower response.

Defender-leader case

If the defender acts as the leader, then for any $d(\cdot) \in \mathcal{D}$ the attacker solves

$$a^*(d) = \arg \max_{a(\cdot) \in \mathcal{A}} J_\alpha(a, d).$$

The defender anticipates this response and solves

$$d^* = \arg \max_{d(\cdot) \in \mathcal{D}} J_d(a^*(d), d).$$

The corresponding equilibrium pair is $(a^*(d^*), d^*)$, and the associated value function becomes

$$V^d(x_0) = \sup_{d(\cdot) \in \mathcal{D}} J_d(a^*(d), d),$$

which represents the maximum cumulative payoff achievable by the defender under optimal follower reaction.

It is important to emphasize that the adopted payoff structure does not correspond to a strict zero-sum game. Although the interaction is adversarial in nature, the attacker and defender objectives are not exact opposites, since $J_\alpha \neq -J_d$. Consequently, the associated Stackelberg value functions $V^\alpha(x_0)$ and $V^d(x_0)$ are defined independently and do not satisfy a sign-symmetry relation.

Summarizing, in either case the Stackelberg game reduces to an individual optimization problem for each agent that can be treated using dynamic programming techniques, subject to a sequential information structure. The dynamic programming formulation is important, since it allows the evaluation of the value of the game (and possibly the optimal strategies) though the Bellman dynamic programming principle, and the corresponding Hamilton-Jacobi-Bellman equation. While in this work we are not solving this equation obtain the value of the game and the optimal policies, still the dynamic programming approach will play a crucial role to the development of the proposed DRL framework, as is analysed below.

3.4 DRL actor-critic approach

While the above formulation defines the Stackelberg equilibrium and the corresponding game values $V^\alpha(x)$ and $V^d(x)$, their analytic computation in closed form is a formidable task, which is often impossible, except for oversimplified cases. Moreover, even the direct numerical approximation in terms of the Bellman or the Hamilton-Jacobi-Bellman equation framework can be a difficult task. To this end, an alternative framework [19, 22, 37, 40] is the DRL approach. This approach still uses the dynamic programming principle as a key step to approaching the problem, but avoids the direct numerical resolution of the resulting Bellman equation (or Hamilton-Jacobi-Bellman). Instead it employs approximations of both the policy functions and the value function by DNNs the parameters of which are tuned so as to approach as close as possible to the Stackelberg equilibrium.

We will explain and analyse the proposed methodology in detail later, but it is important to summarize its main points here before proceeding:

In particular the method uses 3 agents. Two of them, called the actors, correspond to the actual players, i.e. the leader and the follower. The third one, called the critic, is an agent that tries to coordinate the development of the game. Both actors observe the current system state x . In addition, because

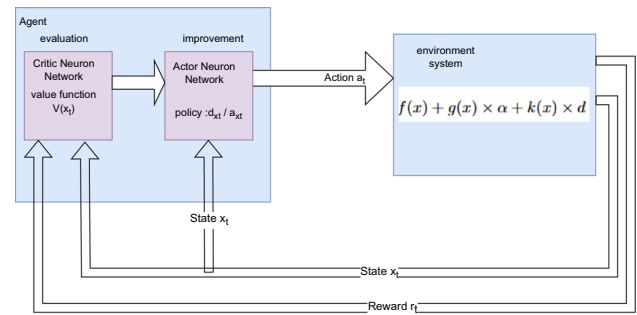


Fig. 3 RL agent interacting with the environment: the actor selects actions based on the policy $\pi(x_t)$, the critic evaluates the value function $V(x_t)$, and the system returns the state x_t and reward r_t

the decision process is sequential, the Follower receives the realized Leader action before selecting its response. Accordingly, the Leader and Follower policies are parameterized as $\pi_L(x; \theta_L)$ and $\pi_F(x, u^L; \theta_F)$, respectively. Neither agent observes the opponent's policy parameters or internal network weights. Based on these actions the state of the system is updated (through (2)) and then judging on the new state of the system, the critic proposes a target value for the game to the actors, in terms of a DNN $\hat{V}_i(x; \phi_i)$, where $i = L, F$, and ϕ_i are the parameters of the corresponding DNN. These target values are approximations to the value of the game and are not arbitrary but based on an update rule which is consistent with the dynamic programming principle that the true value of the game has to obey (see below). After the current approximation of the value of the game by the critic and its announcement to actors, the latter strive to update their policies so as to meet as well as possible their new targets. This procedure is continued until convergence.

We now analyse the above scheme in more detail.

The Actor-Critic architecture illustrated in Fig. 3 consists of two parts: the Actor and the Critic. The first one is the Actor and the second one the Critic. They can be represented by DNNs. The Actor takes decisions for actions. The role of Critic is to 'judge' and evaluate the actions of Actor, the tool to do this is the value function V , when it has the appropriate feedback. [28, 37] After that the strategies that generated by Actor they transferred in Environment. In each iteration the policy can be improved, that lead to the optimal policy.

The DRL framework adapts to both leadership scenarios and time models, optimizing the defense and attack strategies dynamically.

In the implemented algorithm, the quantities

$$\hat{V}^\alpha(x; \phi_\alpha) \quad \text{and} \quad \hat{V}^d(x; \phi_d)$$

denote the outputs of the respective critic neural networks, where ϕ_α and ϕ_d represent the trainable parameters of the attacker and defender critics. These critics provide parametric numerical approximations of the value functions

associated with the previously defined cumulative payoff functionals J_α and J_d .

It is important to emphasize that the algorithm does not perform a direct greedy maximization of the instantaneous reward $r_i(k)$ at each step. Instead, the critic networks approximate a Bellman-consistent value function, while the actor networks—parameterized by θ_α and θ_d —are updated via policy-gradient steps that improve the discounted cumulative payoff over time.

The exact Stackelberg value of the game, defined through the hierarchical maximization problems in the previous subsection, is not computed analytically. Rather, it is approximated numerically through temporal-difference learning. Specifically, the critic networks are trained to satisfy a one-step Bellman consistency relation of the form

$$V_i(x_k) = r_i(k) + \gamma V_i(x_{k+1}),$$

and the actor networks implement successive policy-improvement updates with respect to the learned value functions.

Clearly, in accordance with Remark 1, the current state x_k depends recursively on the previously selected Leader and Follower actions through the system dynamics. Consequently, the Bellman recursion is applied along trajectories that are endogenously generated by the actors' policies, and the learned value function reflects the cumulative effect of these hierarchical control decisions over time.

Therefore, the proposed learning procedure constitutes a recursive dynamic optimization scheme that approximates the hierarchical Stackelberg equilibrium, rather than an explicit closed-form maximization of J_α and J_d .

The value function $V_i(x)$ does not correspond to an analytically computed solution of the Stackelberg maximization problem. Instead, it represents a learned parametric approximation of the discounted cumulative payoff associated with J_i .

In the proposed deterministic setting, the system dynamics and the policies are deterministic mappings. For the attacker-leader configuration, they take the form

$$a_k = \pi_\alpha^L(x_k; \theta_\alpha), \quad d_k = \pi_d^F(x_k, a_k; \theta_d).$$

For the defender-leader configuration, the corresponding order is reversed:

$$d_k = \pi_d^L(x_k; \theta_d), \quad a_k = \pi_\alpha^F(x_k, d_k; \theta_\alpha).$$

No exogenous stochastic disturbances are introduced in the model. Accordingly, the value function is defined without

invoking a probability measure as

$$V_i(x_k) = \sum_{t=k}^{\infty} \gamma^{t-k} r_i(t),$$

where $\gamma \in (0, 1)$ is the discrete-time discount factor.

In this context, π denotes a deterministic policy function mapping the system state to control actions, and not a probability measure. Since no stochastic uncertainty is modeled, no expectation operator is required in the definition of $V_i(x_k)$.

The critic networks approximate this discounted cumulative payoff via temporal-difference updates. The maximization inherent in the Stackelberg formulation is implemented numerically through policy-gradient updates of the actor networks, which adjust the policy parameters so as to improve the critic-estimated long-term value.

Thus, $V_i(x)$ is neither a preset target nor an imposed maximum value. Rather, it is a dynamically learned estimate of the discounted payoff, which—under convergence—approximates the Stackelberg equilibrium value defined in the hierarchical formulation.

Although two leadership scenarios are examined (attacker-first and defender-first), the underlying DRL learning mechanism remains structurally identical. The only difference lies in the sequential order in which actions are computed within each iteration.

Specifically, in the attacker-leader configuration, the attacker policy π_α is evaluated first and the defender responds conditionally through π_d . In the defender-leader configuration, the order is reversed.

Since the training structure, critic updates, Bellman targets, and actor optimization remain identical in both cases, we present a single unified pseudocode formulation using the generic designations *Leader* and *Follower*. The specific role assignment (Leader = attacker or defender) is specified per scenario.

For the attacker-leader scenario, we set Leader = attacker and Follower = defender.

For the defender-leader scenario, we set Leader = defender and Follower = attacker.

Thus, the algorithmic structure remains identical, while the hierarchical information flow is adjusted according to the Stackelberg leadership assignment.

Pseudocode 1 presents a unified Actor–Critic DRL scheme that numerically approximates the hierarchical Stackelberg equilibrium under both leadership scenarios. The algorithm operates in discrete time and follows an online learning structure, where one iteration coincides with one simulation time step.

Initialization The Leader and Follower actors are parameterized by neural networks $\pi_L(x; \theta_L)$ and $\pi_F(x, a_L; \theta_F)$, respectively. These networks implement deterministic state-

Pseudocode 1 Unified Stackelberg Actor–Critic DRL (Leader–Follower Formulation)

```

1: Initialize: Leader actor  $\pi_L(x; \theta_L)$ , Follower actor  $\pi_F(x, a_L; \theta_F)$ ,
   Critic networks  $\hat{V}_L(x; \phi_L)$  and  $\hat{V}_F(x; \phi_F)$ , learning rate  $\alpha$ , discount
   factor  $\gamma$ 
2: for each iteration  $k$  do
3:   Observe state  $x_k$ 
4:   Compute Leader action:

        $a_L(k) = \pi_L(x_k; \theta_L)$ 

5:   Compute Follower action:

        $a_F(k) = \pi_F(x_k, a_L(k); \theta_F)$ 

6:   Update system state:

        $x_{k+1} = f(x_k, a_L(k), a_F(k))$ 

7:   Compute instantaneous rewards  $r_L(k)$  and  $r_F(k)$ 
8:   Critic targets (Bellman update):

        $y_L = r_L(k) + \gamma \hat{V}_L(x_{k+1}; \phi_L)$ 
        $y_F = r_F(k) + \gamma \hat{V}_F(x_{k+1}; \phi_F)$ 

9:   Critic losses:

        $\mathcal{L}_L^{\text{critic}} = (y_L - \hat{V}_L(x_k; \phi_L))^2$ 
        $\mathcal{L}_F^{\text{critic}} = (y_F - \hat{V}_F(x_k; \phi_F))^2$ 

10:  Update  $\phi_L$  and  $\phi_F$  via gradient descent
11:  Actor objectives (policy improvement):

        $\mathcal{L}_L^{\text{actor}} = -\hat{V}_L(x_k, \pi_L(x_k; \theta_L); \phi_L)$ 
        $\mathcal{L}_F^{\text{actor}} = -\hat{V}_F(x_k, \pi_L(x_k; \theta_L), \pi_F(x_k, a_L; \theta_F); \phi_F)$ 

       Update  $\theta_L$  using  $\nabla_{\theta_L} \mathcal{L}_L^{\text{actor}}$ 
       Update  $\theta_F$  using  $\nabla_{\theta_F} \mathcal{L}_F^{\text{actor}}$ 
12: end for

```

feedback policies. The critic networks $\hat{V}_L(x; \phi_L)$ and $\hat{V}_F(x; \phi_F)$ provide parametric approximations of the corresponding discounted value functions. The parameters θ_L, θ_F denote actor weights, while ϕ_L, ϕ_F denote critic weights. The discount factor $\gamma \in (0, 1)$ is consistent with the discretization of the continuous-time functional ($\gamma = e^{-\rho \Delta t}$).

State observation At each iteration k , the current system state x_k is observed. In accordance with Remark 1, the state implicitly depends on previously applied Leader and Follower actions through the system dynamics. Since the system evolution is Markovian, the associated optimization problem admits a dynamic programming formulation, with the value function representing the cost-to-go from state x_k .

In the implemented DRL framework, both agents observe the current system state x_k . The Leader selects its action based solely on x_k , while the Follower conditions its deci-

sion on both the state and the realized Leader action, i.e., $(x_k, a_L(k))$. The critic networks evaluate the state–action configuration resulting from this sequential interaction. No agent directly observes the opponent’s policy parameters or internal network structure; interaction occurs exclusively through the system state and the realized control actions. This preserves the hierarchical Stackelberg information structure implemented in the code.

Leader action computation The Leader policy produces

$$a_L(k) = \pi_L(x_k; \theta_L).$$

This reflects the Stackelberg information structure, where the Leader commits to an action based solely on the current state.

In this work, a clear distinction is made between *simulation steps*, *iterations*, and *episodes*. A *simulation step* corresponds to a single time discretization interval of length Δt , during which the system state is updated according to the discretized dynamics and control actions are applied. An *iteration* refers to one online update of the actor and critic parameters performed at each simulation step. An *episode* consists of a sequence of simulation steps over a fixed time horizon and represents one complete attacker–defender interaction rollout.

Accordingly, in the proposed online learning scheme, one iteration coincides with one simulation time step. This strictly online training mechanism enables continuous adaptation of the policies to the evolving system state.

Follower action computation The Follower then computes

$$a_F(k) = \pi_F(x_k, a_L(k); \theta_F),$$

explicitly conditioning on both the current state and the Leader action. This implements the hierarchical best-response structure at the policy level.

System transition The environment updates according to the discretized system dynamics

$$x_{k+1} = f(x_k, a_L(k), a_F(k)).$$

This deterministic transition captures the cyber–physical interaction between attack and defense actions.

Reward computation Instantaneous rewards $r_L(k)$ and $r_F(k)$ are computed from the stage-wise payoff definitions. These rewards correspond to the discretized version of the continuous-time payoff densities $G_i(x, u_i)$.

Critic target (Bellman update) For each agent, a one-step Bellman target is formed:

$$y_i = r_i(k) + \gamma \hat{V}_i(x_{k+1}; \phi_i).$$

This implements a discounted temporal-difference approximation of the value function fixed point.

Critic update The critic loss

$$\mathcal{L}_i^{\text{critic}} = (y_i - \hat{V}_i(x_k; \phi_i))^2$$

is minimized via gradient descent with respect to ϕ_i . This step enforces Bellman consistency and drives $\hat{V}_i$ toward the fixed point of the discretized Bellman operator associated with the hierarchical game.

Actor objective (policy improvement) The actor losses are defined as

$$\mathcal{L}_L^{\text{actor}} = -\hat{V}_L(x_k, \pi_L(x_k; \theta_L); \phi_L),$$

$$\mathcal{L}_F^{\text{actor}} = -\hat{V}_F(x_k, \pi_L(x_k; \theta_L), \pi_F(x_k, a_L(k); \theta_F); \phi_F).$$

These expressions explicitly encode the chain-rule dependence of the value function on the policy parameters. The Leader actor is optimized with respect to its own policy parameters θ_L , while the Follower actor is optimized conditionally on the Leader action. This preserves the hierarchical Stackelberg structure within the gradient flow.

Actor update Gradient descent is performed on θ_L and θ_F using the respective policy gradients. Importantly, the algorithm does not maximize the instantaneous reward $r_i(k)$ greedily. Instead, it improves the discounted cumulative payoff as approximated by the critic networks.

Policy-gradient interpretation The actor parameter updates are performed via gradient descent on the actor loss functions. Since these losses are defined as the negative of the critic-estimated value functions, minimizing them corresponds to maximizing the discounted cumulative payoff.

For the Leader, the update direction is obtained through the chain rule:

$$\nabla_{\theta_L} \mathcal{L}_L^{\text{actor}} = -\nabla_{\theta_L} \hat{V}_L(x_k, \pi_L(x_k; \theta_L); \phi_L) = -\frac{\partial \hat{V}_L}{\partial a_L} \frac{\partial \pi_L}{\partial \theta_L}.$$

Thus, the Leader policy parameters are adjusted in the direction that increases the critic-estimated long-term value.

Similarly, for the Follower,

$$\nabla_{\theta_F} \mathcal{L}_F^{\text{actor}} = -\frac{\partial \hat{V}_F}{\partial a_F} \frac{\partial \pi_F}{\partial \theta_F},$$

where the value function depends hierarchically on both the state and the Leader action.

Therefore, the actor updates implement a policy-improvement step consistent with approximate dynamic programming and preserve the Stackelberg information structure.

Stackelberg consistency The exact Stackelberg equilibrium, defined via the bilevel optimization problems in the theoretical formulation, is not computed analytically. Instead, the coupled actor-critic updates implement a recursive dynamic optimization procedure that approximates the hierarchical maximization through successive policy-improvement steps.

Leadership assignment The same algorithmic structure applies to both attacker-leader and defender-leader scenarios. Only the designation of Leader and Follower changes. Thus, the DRL training mechanism remains structurally identical, while the information flow reflects the chosen Stackelberg hierarchy.

Overall, Pseudocode 1 constitutes a Bellman-consistent, hierarchy-preserving, actor-critic implementation of the Stackelberg differential game in discretized time.

The proposed Actor-Critic scheme can be interpreted as an instance of approximate dynamic programming (ADP) applied to a hierarchical Stackelberg differential game.

In classical dynamic programming, the value function satisfies the Bellman optimality equation, which characterizes the fixed point of a contraction mapping in the space of value functions. In discrete time, this fixed point is defined by

$$V_i(x_k) = \max_{u_i} [r_i(k) + \gamma V_i(x_{k+1})].$$

In the hierarchical Stackelberg setting, this maximization is structured sequentially: the Leader optimizes anticipating the Follower's best response. Consequently, the Bellman operator becomes a bilevel operator reflecting the Stackelberg information structure.

Since the exact value function cannot be computed analytically for nonlinear dynamics and high-dimensional state spaces, the critic neural networks implement a parametric approximation of the Bellman fixed point. The temporal-difference (TD) update enforces consistency with the discretized Bellman equation, thereby approximating the dynamic programming recursion.

Simultaneously, the actor update corresponds to a policy-improvement step, where the policy parameters are adjusted in the direction that increases the approximated value function. This mirrors classical policy iteration:

- **Policy evaluation:** performed by the critic via TD learning,
- **Policy improvement:** performed by the actor via policy-gradient updates.

Therefore, the proposed algorithm constitutes a recursive hierarchical policy-iteration scheme, where dynamic programming is not solved explicitly but is approximated numerically through function approximation.

Under suitable regularity conditions (bounded rewards, $\gamma \in (0, 1)$, sufficient function-approximation capacity), the critic updates approximate the fixed point of the associated Bellman operator, and the coupled actor-critic dynamics approximate a Stackelberg equilibrium of the underlying differential game.

Role of each component in the Hierarchical learning loop

The learning mechanism consists of four interacting components: the Leader actor, the Follower actor, the critic networks, and the system dynamics.

Leader actor The Leader actor $\pi_L(x_k; \theta_L)$ computes the primary control action based solely on the observed state. Its parameters θ_L are adjusted so as to increase the critic-estimated discounted cumulative payoff. The Leader therefore implements the upper-level optimization step of the Stackelberg game in a parametric and recursive manner.

Follower actor The Follower actor $\pi_F(x_k, a_L(k); \theta_F)$ computes a conditional best-response action. Its update is performed with respect to a value function that explicitly depends on both the state and the Leader action. Thus, the Follower implements the lower-level optimization problem of the Stackelberg hierarchy.

Critic networks Each critic $\hat{V}_i(x; \phi_i)$ approximates the discounted cumulative payoff associated with the corresponding agent. The critic performs a policy-evaluation step by enforcing Bellman consistency through temporal-difference updates. Importantly, the critic does not optimize actions; it evaluates the long-term effect of the actions selected by the actors.

System dynamics The environment implements the deterministic state transition $x_{k+1} = f(x_k, a_L(k), a_F(k))$. It provides the coupling mechanism between attacker and defender, since each agent influences the future state evolution through its control input.

Consequently, the learning process alternates between:

- Value approximation (critic update),
- Policy improvement (actor update),
- State transition through the physical system dynamics.

This recursive interaction realizes a hierarchy-preserving approximate dynamic programming scheme, where the critic approximates the Bellman fixed point and the actors perform successive policy-improvement steps consistent with the Stackelberg information structure.

The Fig. 4 illustrates an Actor-Critic neural network architecture, representing the interactions between attackers and defenders within a system.

The diagram begins with two main inputs, Delta (δ) and Omega (ω), which serve as input variables. These inputs may represent the system's state or external factors that influence the actions of the attacker and defender.

Each side, both the attacker and the defender, has a hidden layer with ReLU (Rectified Linear Unit) activation. The Attacker Hidden Layer (ReLU) receives inputs from Delta and Omega, generating a state representation for the attacker. Similarly, the Defender Hidden Layer (ReLU) also receives inputs from Delta and Omega and generates a state representation for the defender.

The actions are then determined by the Actor networks. The Attacker Actor determines the attacker's action and is labeled as the "Leader," indicating that the attacker acts first. The Defender Actor considers the action of the attacker (Leader) and acts as the "Follower," deciding on the appropriate response.

Both sides have their own Critic layer, which evaluates the value of their actions. The Attacker Critic Hidden Layer (ReLU) and Defender Critic Hidden Layer (ReLU) receive inputs from the hidden layers and the outputs of the actions. These layers calculate the value of each action, using outputs from the Critic layers.

The Critic outputs for each side, Attacker Critic Output (Value) and Defender Critic Output (Value), produce values representing the evaluation of the state for each side, based on their actions.

Based on the attacker's action, an Updated State (Attacker Action) is generated, updating the system's state. This update is then taken into account by the defender for its own action.

In this diagram, it is evident that a total of four neural networks (NNs) are utilized, with two for each player (attacker and defender). For the attacker, the Attacker Actor NN is responsible for generating the attacker's action, selecting the optimal strategy as the Leader, and the Attacker Critic NN evaluates the value of the state resulting from the attacker's action, helping the Actor improve its strategies. For the defender, the Defender Actor NN determines the defender's action as the Follower, responding to the attacker's action, while the Defender Critic NN evaluates the state value resulting from the defender's action, providing feedback on its performance (Fig. 5).

3.5 Evaluation metrics

For each leadership scenario, the following metrics are evaluated in both continuous and discrete time dynamics of a CPS. The following metrics are used to assess the effectiveness of defender and attacker strategies. These metrics enable the evaluation of system stability and resilience against adversarial actions:

- **Resilience:** Measures the system's ability to withstand disturbances caused by the attacker. This metric is calculated as the negative square of the frequency deviation $\omega(t)$, reflecting the system's capability to remain stable despite disruptions.

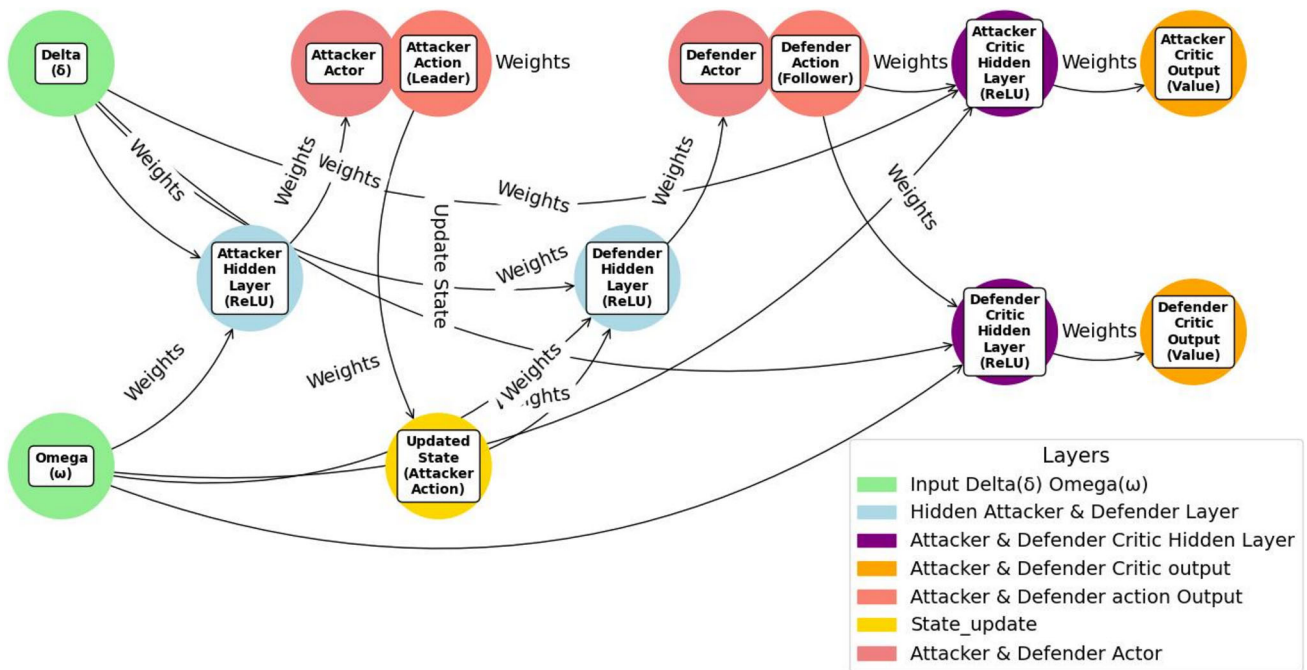


Fig. 4 Schematic representation of the Stackelberg-based Actor-Critic neural network architecture with sequential response, where the attacker acts as the leader and the defender follows. The diagram illustrates the

information flow between actor and critic networks, policy updates, and state transitions under the attacker-first configuration

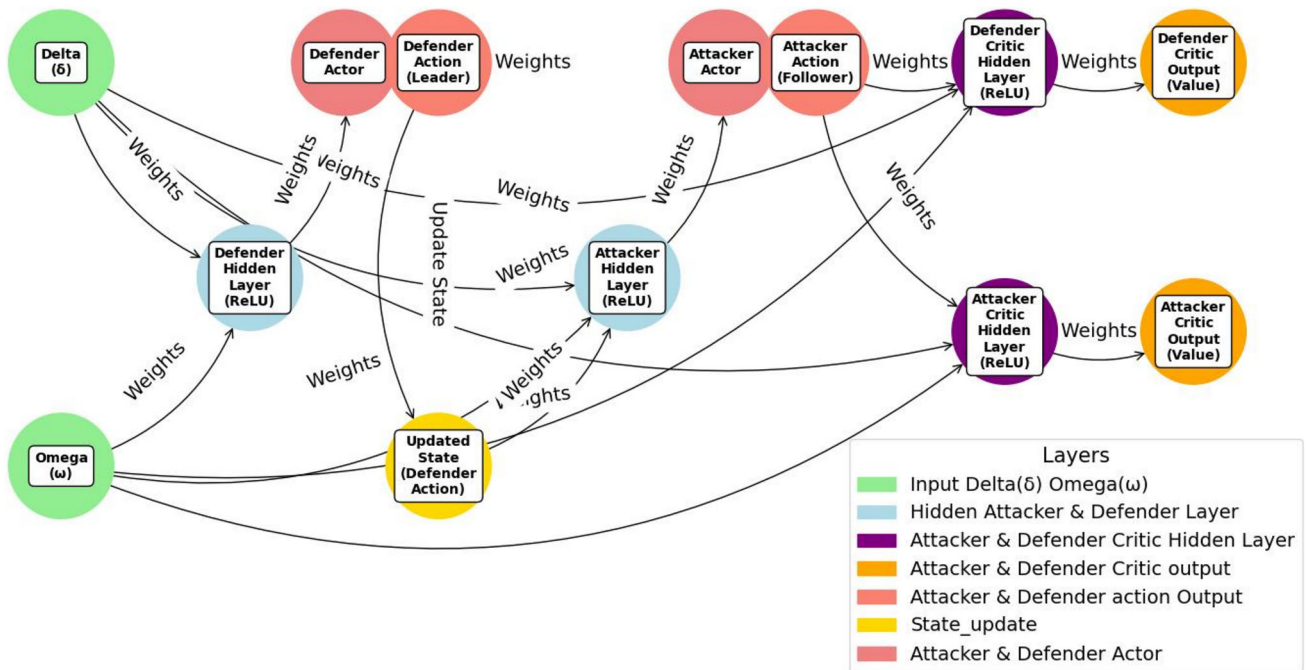


Fig. 5 Schematic representation of the Stackelberg-based Actor-Critic neural network architecture with sequential response, where the defender acts as the leader and the attacker follows. The figure highlights

the corresponding information flow, policy updates, and state transitions under the defender-first configuration

- **Response time:** Defined as the absolute difference between the defender's and attacker's actions, this metric provides insight into the agility and responsiveness of the defender relative to the attacker. Lower values indicate faster defender responses, which are crucial for mitigating the impact of the attack.
- **System damage:** Calculated as the square of the frequency deviation $\omega(t)^2$, this metric represents the level of disturbance experienced by the CPS due to the attacker's actions. High values indicate significant destabilization, while low values suggest successful stabilization efforts by the defender.
- **Cost efficiency:** This metric is determined as the ratio of the defender's action d to the incurred system damage, with adjustments for cases where system damage is zero. It reflects the efficiency of the defender's resource allocation in mitigating damage, with higher values indicating cost-effective strategies.

Although both metrics are computed from the squared frequency deviation, they serve distinct conceptual roles. System Damage quantifies the physical severity of instability as a degradation indicator, whereas Resilience captures the system's recovery capability and stabilization performance. The opposite sign is therefore intentional and reflects degradation versus recovery interpretation rather than a mathematical inconsistency. These metrics are used to evaluate the real-time performance of the CPS under attack and defense actions, facilitating the tuning and validation of policies that enhance system resilience. Stability guarantees under rational jamming attacks were analytically derived in [44], where regional exponential stability and performance bounds were established for linear discrete-time systems under constrained adversarial behavior. In contrast, the present study does not rely on closed-form stability proofs but evaluates resilience metrics empirically under nonlinear dynamics and learning-driven policy adaptation. This distinction reflects the transition from model-based stability certification to data-driven and adaptive security evaluation in complex CPS environments.

The physical basis of the evaluation metrics derives from established power system stability analysis, where frequency deviation and its squared magnitude are commonly used indicators of instability severity. However, the structured combination of these quantities into distinct resilience, damage, response, and cost-efficiency indicators within a dual-role Stackelberg–DRL framework represents an application-specific adaptation introduced in this study for comparative evaluation across discrete and continuous time formulations.

4 Simulation

This section presents a detailed overview of the simulation process, including the tools, methodologies, and key observations from the results. Simulations are conducted to validate the effectiveness of dynamic control strategies in CPS using both discrete and continuous time models. This approach is designed to demonstrate the impact of leadership roles on system resilience and overall defense strategy.

4.1 Implementation framework

The simulations are developed using Python, with TensorFlow as the primary framework for constructing the DRL models. This setup ensures efficient handling of the actor-critic algorithm, which forms the core of the simulation process. The models are trained to adaptively respond to potential threats and optimize defense mechanisms based on real-time data.

4.2 System constants and functions

The simulations involve the following system constants:

- $M = 10.0$: Inertia
- $D = 1.0$: Damping
- $P_m = 1.0$: Mechanical power
- $P_{\max} = 1.5$: Maximum electrical power
- $dt = 0.01$: Time step

As detailed in Sections 3.3.1 and 3.3.2, the system state x is defined as comprising the rotor angle δ and the frequency deviation ω , while the natural dynamics of the power grid CPS are characterized by the function $f(x)$. This function encapsulates the generator's behavior based on key parameters such as the mechanical power P_m , the maximum electrical power $P_{\max}$, the damping coefficient D , and the inertia constant M . In addition, the influences of the attacker's and defender's control actions are represented by the functions $g(x)$ and $k(x)$, respectively, which scale their respective inputs by the system's inertia. These definitions provide the foundation for analyzing the continuous-time dynamics of the power grid CPS, and further details can be found in Sections 3.3.1 and 3.3.2.

These functions are constructed based on the system's physical properties and the interactions outlined in the code. The function represents the natural dynamics of the system, including the effects of inertia and damping, while and denote the influence of control inputs and attack actions, respectively. These functions have been derived based on the work by Peter W. Sauer and M. A. Pai in the book "Power System Dynamics and Stability" [31]. They describe the inherent system dynamics $f(x)$, the control influence by the defender

$g(x)$, and the impact of the attack actions $k(x)$. These functions, along with the constants, are essential for modeling the dynamic response of the CPS under adversarial conditions.

4.3 Key variables: ω and δ

The variables ω (rotor frequency) and δ (rotor angle) are selected due to their fundamental role in representing the dynamical stability of the power grid. In particular, the state vector is defined as $x = (\delta, \omega)^T$, and both variables evolve according to the system dynamics in (1).

In accordance with Remark 1, the trajectories $\omega(t)$ and $\delta(t)$ depend implicitly on the applied control actions $a(t)$ and $d(t)$. Therefore, cyberattacks and defensive responses influence the system exclusively through their effect on these state variables.

The variable ω reflects frequency deviations in the system, serving as an indicator of disturbances and of the immediate dynamical impact of cyberattacks on the physical process. The variable δ represents the angular position of the rotor, which is essential for maintaining synchronization within the interconnected power grid.

Cyberattacks may disrupt the balance between mechanical and electrical power, causing deviations in ω and δ . Sustained or growing deviations in ω indicate potential instability and may lead to cascading failures if not properly mitigated.

The defender's role is therefore to influence the state evolution through the control input $d(t)$, counteracting the destabilizing effect of the attacker input $a(t)$. Stability preservation is thus formulated as a control problem on the state variables (δ, ω) , which serve as the measurable interface between cyber actions and physical system behavior.

4.4 Discrete time model

In the discrete time simulation, the environment is represented as a step-by-step process where the interactions between the attacker and defender are examined at each time interval. Key variables include system state transitions, attack actions, defense responses, and overall system stability. The discrete time model focuses on understanding how incremental decision-making affects system behavior and resilience under different conditions.

4.5 Continuous time model

The continuous time simulation uses a more fluid approach to model interactions, represented through differential equations. This allows for evaluating system responses in a more natural and uninterrupted manner. As defined earlier (see Eq. (1)), the system dynamics are governed by the natural dynamics $f(x)$ combined with the attack input $g(x) \cdot \alpha$ and the defense input $k(x) \cdot d$. These functions are used to

simulate the system's behavior, highlighting the influence of attack and defense strategies on overall dynamics. The system dynamics are numerically integrated using a fourth-order Runge–Kutta scheme, as implemented in standard scientific computing programmatic implementation, to ensure accurate and reproducible simulations.

Solving differential equations in continuous time

In this study, the numerical solution of the system's differential equations was achieved using the Runge–Kutta method. The Runge–Kutta method is a widely used approach for solving ordinary differential equations (ODEs) numerically, offering high accuracy without requiring complex computations.

Specifically, the fourth-order Runge–Kutta method was employed to integrate the system's dynamic equations. For each time step Δt , the state values were updated using the following relations:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{\Delta t}{2}, y_n + \frac{\Delta t}{2} k_1\right), \\ k_3 &= f\left(t_n + \frac{\Delta t}{2}, y_n + \frac{\Delta t}{2} k_2\right), \\ k_4 &= f(t_n + \Delta t, y_n + \Delta t \cdot k_3), \\ y_{n+1} &= y_n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4), \end{aligned}$$

where y_n represents the state of the system at time t_n , and $f(t, y)$ is the function describing the system's dynamics.

The Runge–Kutta method was chosen for its accuracy in modeling dynamic systems, particularly in continuous-time applications [3, 15]. This ensures stability and precision in numerical simulations, especially for problems where the linearity and simplicity of the equations are critical.

In continuous-time reinforcement learning, iterations serve as a practical proxy for tracking system dynamics and agent performance. While continuous time (t) is inherently embedded in the simulation through a discretization step (Δt), tracking iterations allows for a clearer representation of progress in the training process. This approach aligns with prior work, such as Ramstedt and Pal [30], where iterations are used to monitor learning in numerically simulated continuous systems without introducing the complexity of mapping to real-world time. Additionally, Doya [7] emphasizes that in continuous-time systems, the discretization of time required for numerical integration inherently makes iterations a natural and stable proxy for tracking progress. He notes that iterations provide a more interpretable representation of the learning process by abstracting away the complexities of real-time measurements, enabling a direct comparison of system performance across different states and steps.

To enhance the interpretability of the results, the x-axis in the visualizations has been mapped to real time (t) using the relation $t = \text{iteration} \times \Delta t$, where Δt represents the simulation time step. This approach directly links the simulation dynamics to real-world time, offering a straightforward representation of system behavior over time.

4.6 Evaluation metrics

To quantify the impact of leadership roles on system resilience, two main metrics are used: resilience score and system damage. These metrics assess the system's ability to maintain functionality during and after an attack and the extent of damage inflicted during adversarial interactions. The results are presented through graphical outputs to highlight differences in system behavior when the attacker or defender assumes the leadership role.

In conclusion, the simulations emphasize the importance of adaptive strategies in CPS, highlighting the potential benefits of a proactive defense framework. This section provides a comprehensive overview, linking theoretical concepts to practical implementation and results.

4.7 Convergence and early stopping

This subsection complements the simulation results presented in the previous sections by examining whether the reported numerical outcomes correspond to stable and convergent learning behavior. A critical concern in reinforcement learning-based security frameworks for CPS is the reliability and trustworthiness of the learned strategies. In particular, readers must be able to assess whether the reported figures and performance metrics reflect stable and near-optimal solutions, rather than transient behaviors arising from incomplete training or overfitting.

The proposed learning architecture is based on an Actor-Critic formulation for both attacker and defender agents, operating under a leader-follower (Stackelberg) game structure. Due to the nonlinear system dynamics and the use of deep neural networks for policy and value function approximation, deriving a closed-form analytical proof of convergence to a sub-game perfect equilibrium is generally intractable. This limitation is well known in DRL applied to continuous-state CPS. Consequently, convergence is assessed empirically using multiple complementary indicators.

Training stability is first evaluated through the evolution of the Critic loss functions for both agents. As training progresses, the Critic networks approximate the underlying value functions associated with the Stackelberg interaction. Convergence is indicated by the stabilization of the Critic losses around bounded values, without persistent oscillations or divergence. This behavior suggests that temporal-

difference errors decrease over time and that the value estimates become self-consistent across training iterations.

Convergence is further examined at the policy level by monitoring the evolution of the Actor outputs. After an initial exploration phase, both attacker and defender policies exhibit progressively smoother trajectories and reduced variance, indicating a transition from exploration to exploitation. The resulting control actions remain stable across episodes, providing additional evidence that the learning process converges to a steady behavioral regime rather than exhibiting unstable or chaotic dynamics.

To operationalize the convergence and stability criteria discussed above, an early stopping mechanism is incorporated into the training process. This mechanism ensures that training terminates once performance improvements saturate, thereby preventing overfitting and unnecessary computation while preserving generalization.

In the proposed continuous-time simulations, the actor and critic networks are updated in a strictly online manner, with one parameter update performed at each simulation time step.

Strictly online parameter updates may induce oscillatory behavior in continuous-time training due to increased sensitivity to high-frequency noise.

While this design choice enables rapid adaptation to evolving system dynamics, it may also amplify high-frequency fluctuations in the learned control signals, a phenomenon commonly referred to as chattering.

This behavior is particularly pronounced in continuous-time settings, where frequent updates combined with nonlinear dynamics can lead to oscillatory policy adjustments. To mitigate these effects and ensure reliable performance evaluation, an early stopping mechanism is employed, terminating training once the critic losses and control actions stabilize. Consequently, the reported results correspond to stabilized policy regimes rather than transient high-frequency fluctuations.

4.7.1 Early stopping mechanism

The early stopping mechanism employed in our implementation is designed to terminate training before completing all iterations when the model's performance ceases to improve over a predefined window. Performance is monitored using a joint instantaneous reward-based performance signal derived from the attacker and defender objectives., which reflect the quality of the learned attacker and defender strategies within the CPS environment. The description of the early stopping mechanism was extended to clarify that the stopping criterion is based on a combined cumulative performance signal defined as the difference between the attacker's and defender's rewards. This joint metric is used to monitor stabilization of the overall Stackelberg interaction, ensuring that

training terminates when the joint attacker–defender performance signal stabilizes and no further improvement is observed.

Early-stopping reward signal To define a single scalar criterion for early stopping in the attacker–defender setting, we monitor a combined performance signal constructed from the two agents' rewards. Let ω_{t+1} denote the frequency deviation at the next step and d_t the defender action. In our implementation, the attacker and defender instantaneous rewards are defined as

$$r_A(t) = \omega_{t+1}^2,$$

$$r_D(t) = -\omega_{t+1}^2 + \lambda |d_t|,$$

with $\lambda = 0.1$. The early-stopping signal is then given by

$$r_{\text{stop}}(t) = r_A(t) - r_D(t) = 2\omega_{t+1}^2 - \lambda |d_t|.$$

We monitor this signal during training and apply a patience-based criterion: training is terminated if the monitored performance signal does not improve beyond the best observed value for a predefined number of consecutive iterations. The monitored signal corresponds to the instantaneous reward difference evaluated at each training iteration. The stopping decision is based on the best observed value of this signal across iterations.

During training, the monitored joint reward signal at each iteration is compared against the best performance observed so far. When an improvement is detected, the best reward value is updated and a patience counter is reset. If no improvement is observed for a predefined number of consecutive iterations, the patience counter increases. Training is halted once this counter reaches a specified threshold, indicating that further training is unlikely to yield meaningful performance gains.

Early stopping serves multiple purposes within the proposed framework. First, it reduces the risk of overfitting by preventing the policies from adapting excessively to the training trajectories. Second, it improves computational efficiency by avoiding unnecessary iterations once performance plateaus. Finally, by terminating training at the point of maximal observed performance, early stopping enhances the generalization capability of the learned policies. The use of early stopping as a regularization and convergence control mechanism is well established in the machine learning literature [14], and is commonly adopted in reinforcement learning to ensure stable training behavior [32].

Overall, the combination of Critic loss stabilization, policy behavior consistency, and early stopping provides empirical evidence that the learned strategies converge to stable and near-optimal solutions under the given modeling assumptions. Although strict analytical convergence guarantees to

a sub-game perfect equilibrium are not available for DRL in nonlinear CPS, the observed convergence behavior aligns with best-response stability in Stackelberg games. From a game-theoretic perspective, as training progresses, the follower adapts consistently to the leader's strategy, and both agents settle into stable policies that reflect a locally consistent leader–follower equilibrium under the learned value functions.

5 Results and analysis

The results illustrate how different leadership configurations, whether attacker-led or defender-led, influence system resilience and response strategy. Graphs and data visualizations compare the performance of both discrete and continuous models, showcasing how leadership changes can lead to significant outcome variations. In this analysis, the columns **critic losses**, **final metrics**, and **basic numerical values** from Table 1 were utilized for each scenario. Specifically:

- **Basic numerical values** The values of δ (angle deviation) and ω (frequency deviation) were used to describe the system's response to each situation. The final actions of the attacker (a) and the defender (d) were analyzed to evaluate the intensity of their strategies in each scenario.
- **Critic losses** The final Critic loss values for the attacker and the defender were interpreted as indicators of the effectiveness of neural networks in implementing their policies. Additionally, the average, maximum, and minimum losses were analyzed to assess the computational burden on the system during the interaction.
- **Final metrics** The **resilience** values were analyzed as key indicators of the system's ability to absorb and recover from disruptions. The **system damage** explained the overall level of harm sustained by the CPS in each scenario. The **response time** and **cost efficiency** were evaluated to assess the timeliness and economic feasibility of the strategies.

5.1 Scenario 1: defender acts as leader (continuous time)

In this scenario, the defender is allowed to act first, preemptively mitigating potential attacks. The results show a high angle deviation of $\delta = 15.2671$ rad and a frequency deviation of $\omega = 2.0121$ rad/s. Both the attacker and the defender maximize their actions, with $a = 1.0$ and $d = 1.0$. The resilience metric is -4.0485 , indicating limited ability to absorb disruptions. The response time is 0.0000 , showing an immediate reaction typical of continuous time systems. The system sustains significant damage of 4.0485 , and the

cost efficiency is -0.2470 , highlighting the higher cost of this defensive approach.

The Critic losses show that the final loss for the attacker is 0.0728 , while the defender's loss is 5.2019 . The average losses are 0.9347 for the attacker and 2.3750 for the defender. Maximum Critic losses reach 3.0331 for the attacker and 8.1951 for the defender, while the minimum losses are 0.0000 for both. These findings indicate that while the defender can counteract the attacker effectively, the computational cost is high, particularly for the defender during peak stress moments. The defender should integrate adaptive and real-time predictive strategies to minimize system damage and improve cost efficiency.

5.2 Scenario 2: attacker acts as leader (continuous time)

When the attacker acts first, the system exhibits a smaller angle deviation of $\delta = 0.2533$ rad and a frequency deviation of $\omega = 0.0856$ rad/s. The attacker maximizes their action ($a = -1.0$), while the defender takes no action ($d = 0.0$). Resilience is -0.0073 , reflecting a relatively balanced state of the system. The response time is 1.0000 , indicating a more measured reaction compared to Scenario 1. The system damage is significantly reduced to 0.0073 , while the cost efficiency is -0.0000 , suggesting a balance between cost and effectiveness.

The Critic losses show that the final losses are 0.0000 for both the attacker and defender, with the maximum loss for the attacker being 0.0001 and 0.0000 for the defender. This scenario underscores the potential benefits of allowing the attacker to act first, provided the defender's response is well-optimized.

The defender's strategy should prioritize anticipation and preemptive measures to maintain system stability.

5.3 Scenario 3: defender acts as leader (discrete time)

In discrete time, the proactive strategy of the defender produces improved results. The angle deviation is reduced to $\delta = 0.4207$ rad and the frequency deviation to $\omega = 0.1790$ rad/s. The defender's action reaches $d = 0.2070$, while the attacker's action is minimized to $a = -0.1385$, reflecting the effectiveness of the defensive strategy.

The resilience metric improves significantly to -0.0321 , and the system damage decreases to 0.0321 , representing minimal disruption. The response time increases to 0.3455 due to the discrete decision-making intervals. However, the cost efficiency deteriorates to -6.4583 , highlighting the high cost associated with this defensive approach.

The Critic losses indicate a well-balanced interaction. The final loss for the attacker is 0.0002 , while the defender's

loss is effectively zero at 0.0000 . The average losses for the attacker and defender are 0.0002 and 0.0000 , respectively. The maximum losses are 0.0004 for the attacker and 0.0001 for the defender, demonstrating lower computational strain compared to continuous time scenarios.

Synchronizing defensive actions and optimizing decision-making intervals could further enhance system resilience and cost efficiency, minimizing disruptions and maintaining system stability.

5.4 Scenario 4: attacker acts as leader (discrete time)

When the attacker acts first in discrete time, the system exhibits higher angle deviation ($\delta = 2.5142$ rad) and frequency deviation ($\omega = 0.4561$ rad/s). The attacker maximizes their action ($a = 1.0$), while the defender takes no action ($d = 0.0$). The resilience metric is -0.2080 , and the system damage is 0.2080 , moderate compared to other scenarios. The response time improves to 1.0000 , reflecting faster reactions than in Scenario 3. Cost efficiency reaches -0.0000 , highlighting balanced cost-effectiveness.

The Critic losses show that the final loss for the attacker is 0.0051 , while the defender's final loss is 0.0067 . Maximum Critic losses are 0.0241 for the attacker and 0.0239 for the defender. This scenario highlights the defensive strain when reacting to preemptive attacks by the attacker. Enhancing decision-making algorithms to trigger timely defensive actions could improve the system outcomes.

5.5 Extended analysis of scenarios

The performance of the CPS under varying scenarios provides key insights into the interaction between resilience, system damage, and defensive strategies. A detailed comparison of all metrics across the four scenarios follows.

Angle deviation (δ) and frequency deviation (ω)

Scenario 3 (Defender first in discrete time) achieves the smallest angle and frequency deviations at $\delta = 0.4207$ rad and $\omega = 0.1790$ rad/s. These results highlight the effectiveness of a proactive defensive strategy with structured decision intervals.

Scenario 1 (Defender first in continuous time) exhibits the highest angle deviation at $\delta = 15.2671$ rad and the highest frequency deviation at $\omega = 2.0121$ rad/s. This significant disparity reflects the limitations of continuous time frameworks in containing disruptions despite immediate responses.

Scenario 2 (Attacker first in continuous time) and Scenario 4 (Attacker first in discrete time) fall between these extremes. In Scenario 2, $\delta = 0.2533$ rad and $\omega = 0.0856$ rad/s suggest better system control compared to Scenario 1 but lack the

Table 1 Numerical results comparing dynamic performance and training metrics across continuous- and discrete-time environments for both attacker-first and defender-first scenarios. Metrics include state variables (δ , ω), final action values, critic losses, and system-level indicators (resilience, damage, and cost-efficiency)

Time	Scenario	Basic numerical values	Critic Losses	Final metrics
Continuous	Defender first	δ : 15.2671 rad ω : 2.0121 rad/s Final Action (α): 1.0000 Final Action (d): 1.0000	Attacker: 0.0728 Defender: 5.2019 Avg Attacker: 0.9347 Avg Defender: 2.3750 Max Attacker: 3.0331 Max Defender: 8.1951 Min Attacker: 0.0000 Min Defender: 0.0000	Resilience: -4.0485 Response Time: 0.0000 System Damage: 4.0485 Cost Efficiency: -0.2470
	Attacker first	δ : 0.2533 rad ω : 0.0856 rad/s Final Action (α): -1.0000 Final Action (d): 0.0000	Attacker: 0.0000 Defender: 0.0000 Avg Attacker: 0.0000 Avg Defender: 0.0000 Max Attacker: 0.0001 Max Defender: 0.0000 Min Attacker: 0.0000 Min Defender: 0.0000	Resilience: -0.0073 Response Time: 1.0000 System Damage: 0.0073 Cost Efficiency: -0.0000
Discrete	Defender first	δ : 0.4207 rad ω : 0.1790 rad/s Final Action (α): -0.1385 Final Action (d): 0.2070	Attacker: 0.0002 Defender: 0.0000 Avg Attacker: 0.0002 Avg Defender: 0.0000 Max Attacker: 0.0004 Max Defender: 0.0001 Min Attacker: 0.0000 Min Defender: 0.0000	Resilience: -0.0321 Response Time: 0.3455 System Damage: 0.0321 Cost Efficiency: -6.4583
	Attacker first	δ : 2.5142 rad ω : 0.4561 rad/s Final Action (α): 1.0000 Final Action (d): 0.0000	Attacker: 0.0051 Defender: 0.0067 Avg Attacker: 0.0116 Avg Defender: 0.0106 Max Attacker: 0.0241 Max Defender: 0.0239 Min Attacker: 0.0000 Min Defender: 0.0000	Resilience: -0.2080 Response Time: 1.0000 System Damage: 0.2080 Cost Efficiency: -0.0000

structured benefits of Scenario 3. Scenario 4 shows larger deviations ($\delta = 2.5142$ rad, $\omega = 0.4561$ rad/s) due to the attacker's advantage in acting first.

Analyzing the variance in Critic losses between the attacker and the defender provides information on the computational complexity of different strategies. Scenarios with higher maximum Critic losses, such as Scenario 1, where the defender's loss peaks at 8.1951, indicate moments of significant computational stress. Scenario 3, with minimal maximum Critic losses for the attacker (0.0004) and defender (0.0001), demonstrates the benefits of structured decision-making intervals. Reducing spikes in Critic losses through

more consistent policies could improve system stability and performance.

Resilience and system damage

A detailed analysis of the resilience metrics (Resilience) and system damage (System Damage) metrics reveals an inverse correlation, with scenario 3 achieving the best performance. The resilience of -0.0321 and the system damage of 0.0321 are unmatched, reflecting the robustness of a discrete proactive defense.

Scenario 1 shows the weakest resilience at -4.0485 , accompanied by the highest system damage of 4.0485 . Despite the defender's immediate actions, continuous-time systems fail to mitigate attacks effectively.

Scenario 2 slightly improves on Scenario 1 with a resilience of -0.0073 and system damage of 0.0073 , benefiting from the attacker's preemptive actions. Scenario 4 achieves moderate resilience (-0.2080) and system damage (0.2080) due to the structured response intervals in discrete time, but is less effective than Scenario 3.

Response time

Response time is critical in determining the effectiveness of system actions. Scenarios 1 and 2, both in continuous time, exhibit the shortest response times at 0.0000 and 1.0000 , respectively. However, these immediate responses do not translate into better outcomes, as seen in higher system damage and lower resilience values.

Scenario 3's response time of 0.3455 strikes a balance between quick actions and measured decisions, resulting in superior system performance. Scenario 4 has a response time of 1.0000 , slightly slower but still effective due to the structured intervals.

Cost efficiency

Cost efficiency varies significantly across the scenarios, highlighting the trade-offs between economic and operational metrics. Scenario 4 achieves the most balanced cost efficiency at -0.0000 , indicating minimal resource wastage. However, this comes at the expense of increased system damage and reduced resilience.

The cost efficiency of scenario 3 of -6.4583 reflects the higher costs of implementing a discrete proactive defense. Nevertheless, this approach ensures the best resilience and the lowest system damage, validating the investment.

Scenarios 1 and 2, with cost efficiencies of -0.2470 and -0.0000 , show that continuous-time frameworks incur lower economic costs but fail to provide comparable system protection.

Critic losses

The variance in critic losses provides insight into the computational complexity of each scenario. Scenario 3 exhibits minimal Critic losses for both attacker (final: 0.0002 , max: 0.0004) and defender (final: 0.0000 , max: 0.0001), highlighting the stability of structured decision intervals.

Scenario 1, by contrast, experiences significant critic losses, particularly for the defender (final: 5.2019 , max: 8.1951). This indicates computational strain due to the unstructured and reactive nature of continuous-time systems.

Scenarios 2 and 4 show moderate losses, benefiting from the attacker's first action in each framework.

In the analysis of the scenarios, Scenario 3 (Defender first in discrete time) stands out with the lowest angle deviation ($\delta = 0.4207$ rad), frequency deviation ($\omega = 0.1790$ rad/s), resilience (-0.0321), and system damage (0.0321). Despite higher costs, it achieves unparalleled system protection.

Scenario 1 (Defender first in continuous time) shows the worst performance across almost all metrics, with the highest system damage (4.0485), lowest resilience (-4.0485), and largest Critic losses for the defender.

Scenario 4 (Attacker first in discrete time) achieves balanced cost efficiency (-0.0000) but sacrifices resilience and system damage, which remain significantly worse than Scenario 3.

Discrete-time frameworks (Scenarios 3 and 4) consistently outperform continuous-time (Scenarios 1 and 2) in terms of resilience and system damage, demonstrating the value of structured decision intervals.

Allowing the defender to act first (Scenarios 1 and 3) generally results in better outcomes than attacker-first scenarios, provided the framework supports proactive strategies.

Impact of response time on system damage

The response time (Response time) also plays a critical role in determining system outcomes. In discrete-time scenarios, such as Scenario 3, the response time of 0.3455 coincides with significantly reduced system damage. Conversely, the immediate response times (0.0000) in continuous time scenarios, such as Scenario 1 and Scenario 2, do not prevent higher damage levels. This indicates that while rapid responses are advantageous, the effectiveness of the action taken is paramount.

Cost efficiency trends across scenarios

The Cost efficiency metric highlights notable trade-offs between resource expenditure and system performance. While Scenario 4 achieves a balanced cost efficiency of -0.0000 , this is accompanied by a relatively higher system damage of 0.2080 . In contrast, Scenario 3, despite a poorer cost efficiency of -6.4583 , achieves significantly better resilience and lower damage. This suggests that improving cost efficiency without compromising system resilience could further optimize performance.

Interdependence of final actions and system metrics

The final actions (a and d) of the attacker and defender show strong interdependencies with other system metrics. For example, in Scenario 3, the defender's action ($d = 0.2070$) and the attacker's action ($a = -0.1385$) align with the

best overall performance of the system, including the lowest damage and the highest resilience. This underscores the importance of defensive actions in driving favorable outcomes.

Insights from discrete versus continuous time frameworks

A comparison between discrete and continuous time frameworks reveals that discrete time scenarios tend to distribute system stress more evenly, as seen in the lower maximum Critic losses for both attacker (0.0004) and defender (0.0001) in Scenario 3. Continuous-time scenarios, while offering immediate responses, lead to higher computational and system stress, as reflected in the higher values of System Damage and Critic Losses.

The discrete framework allows for better resilience and reduced system damage, particularly when the defender is proactive. Continuous frameworks, despite faster response times, show a tendency for higher disruptions due to unstructured decision intervals. Future work should explore hybrid approaches to balance immediate responses with structured decision-making.

The analyses presented here reinforce the critical role of resilience, response time, and defensive actions in optimizing CPS performance. By focusing on strategies that balance cost efficiency, reduce system stress, and improve resilience, the effectiveness of defense mechanisms can be significantly improved. These insights pave the way for targeted improvements in both discrete and continuous time frameworks.

5.6 Introduction to diagram analysis

To provide a comprehensive understanding of the system dynamics and interactions between attackers and defenders within (CPS), this section presents a detailed analysis of the key performance metrics derived from the simulation results. Utilizing a series of visual representations, the analysis evaluates the evolution of critical parameters such as resilience, system damage, cost efficiency, rotor angle, and frequency across various scenarios. Each Figure highlights specific aspects of the system's behavior under distinct strategic frameworks, offering insights into the impact of continuous versus discrete-time approaches, as well as attacker-first versus defender-first strategies. The subsequent discussion not only interprets the visual data, but also correlates them with numerical findings to identify optimal strategies for enhancing the robustness and adaptability of CPS against cyber threats.

Here is the detailed analysis of the scenarios presented in Figure 15, focusing on the CPS power grid and the dynamics between the attacker and defender across both continuous and discrete time models:

5.7 Comparative analysis of actions in a CPS power grid

Figure 6 illustrates the interactions between the attacker and the defender in four different scenarios in a CPS power grid. Each subFigure represents a distinct combination of leadership roles and time frameworks, offering insights into system dynamics and stability.

Scenario 1: defender first (continuous time)

Figure 6a depicts the scenario where the defender acts first in a continuous-time framework. The graph shows a rapid increase in both attacker and defender actions, with both stabilizing at their maximum values ($a = 1.0$, $d = 1.0$) within the first 500 iterations. The synchronized rise of actions highlights an escalation phase driven by adaptive best-response learning rather than a purely reactive response, where the defender's preemptive measures force the attacker to intensify its actions. As the attacker increases its action magnitude to amplify system disruption, the defender correspondingly intensifies its control effort to preserve stability, resulting in a coupled feedback loop that drives both actions toward their saturation limits.

However, the continuous time framework lacks structured decision intervals, leading to overshooting behavior and computational inefficiency. The steep slope and convergence of actions indicate high system strain, which aligns with the significant resilience loss (-4.0485) and system damage (4.0485) observed in this scenario.

While immediate responses can mitigate initial disruptions, the absence of structured intervals results in oscillatory dynamics and high system stress, reducing overall system performance.

Scenario 2: attacker first (continuous time)

Figure 6b presents the scenario in which the attacker takes the lead in continuous time. The attacker's action steadily decreases to $a = -1.0$ by iteration 600, while the defender remains inactive ($d = 0.0$). This reflects the dominance of the attacker in exploiting system vulnerabilities without encountering significant resistance.

Despite the aggressive approach of the attacker, the structured decrease in their action leads to minimal system disruptions, as reflected in the low resilience loss (-0.0073) and system damage (0.0073). The continuous time framework, however, limits the defender's ability to adapt in real time, as evidenced by the lack of a significant response.

The dominance of the attacker highlights the importance of dynamic defensive strategies to counteract pre-emptive attacks effectively.

Scenario 3: defender first (discrete time)

Figure 6c illustrates the scenario in which the defender acts first in a discrete time framework. The defender's action ($d = 0.2070$) and the attacker's action ($a = -0.1385$) stabilize after initial oscillations, demonstrating the structured

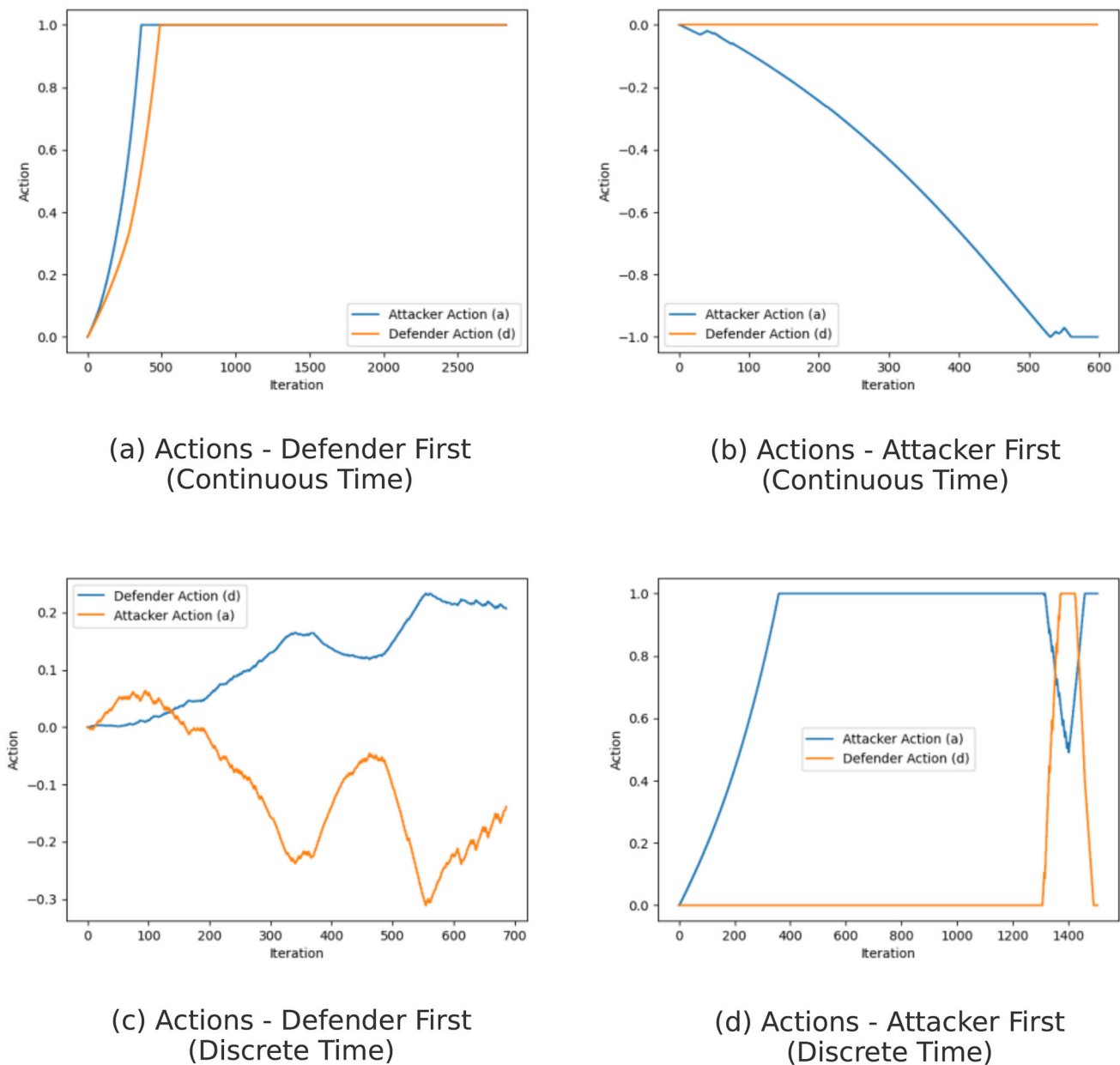


Fig. 6 Comparative analysis of attacker and defender actions across different leadership scenarios

and controlled responses enabled by discrete time decision intervals.

This scenario achieves the best overall system performance, with the lowest angle deviation ($\delta = 0.4207$ rad) and frequency deviation ($\omega = 0.1790$ rad/s). The structured nature of discrete time minimizes computational strain and ensures that the defender's proactive measures effectively counteract the attacker's influence. These dynamics are validated by the highest resilience (-0.0321) and the lowest system damage (0.0321) among all scenarios.

Structured intervals of discrete time provide a significant advantage in stabilizing system dynamics and ensuring robust defense mechanisms.

Scenario 4: attacker first (discrete time)

Figure 6d shows the scenario in which the attacker leads in a discrete-time framework. The attacker's action ($a = 1.0$) quickly stabilizes, while the defender follows with a delayed response ($d = 1.0$), indicating a reactive strategy. This delay allows the attacker to maintain an initial advantage, causing moderate system disruptions.

Although the discrete-time framework prevents oscillatory behavior and ensures structured decision making, the delay in defensive action leads to moderate resilience loss (-0.2080) and system damage (0.2080). The plot highlights the critical importance of timing and synchronization between attacker and defender actions. The balance between cost efficiency (-0.0000) and moderate performance results reflects the trade-offs inherent in reactive defensive strategies.

The defender's delayed response emphasizes the need for faster decision-making algorithms to reduce the attacker's advantage and enhance system protection.

The leadership role (attacker-first vs. defender-first) significantly impacts system outcomes. Defender-first strategies, particularly in discrete time, consistently outperform attacker-first scenarios in terms of resilience and system stability.

Figure 6 visually shows that discrete-time frameworks, combined with proactive defensive strategies, offer the most robust protection for CPS power grids. Investing in structured decision making and reducing defensive delays can further optimize resilience and system performance under attack.

5.8 Analysis of resilience across different scenarios

Figure 7 presents the resilience trends for four different scenarios in a CPS power grid. These scenarios are categorized by leadership roles (defender first or attacker first) and temporal frameworks (continuous or discrete time). The resilience metric, which represents the degradation and recovery of system performance, highlights the impact of attack and defense strategies on system stability over iterations.

In the defender-first continuous time scenario (Fig. 7a), the resilience metric exhibits a steady decline over 1400 iterations, ultimately reaching -4.0 . This significant degradation indicates the system's inability to stabilize due to the lack of structured decision intervals in continuous time. Despite the preemptive role of the defender, the unstructured nature of continuous time fails to effectively mitigate the impact of disruptions, leading to considerable system strain.

When the attacker leads in a continuous time framework (Fig. 7b), the resilience initially declines sharply, reaching its lowest point (-0.012) around iteration 150. Following this decline, the system exhibits partial recovery, with resilience slightly improving by iteration 300. This behavior reflects the attacker's early dominance in exploiting vulnerabilities, while the defender's reactive measures achieve limited adaptation over time. The inability of the continuous-time framework to provide structured responses hampers overall system performance.

In the defender-first discrete time scenario (Fig. 7c), the resilience metric shows a gradual and controlled decline, stabilizing at approximately -0.03 after 700 iterations. The

structured decision-making intervals of discrete time enable the defender to execute proactive strategies effectively, minimizing the degradation of system performance. This scenario highlights the advantages of discrete-time frameworks in maintaining stability and resilience against attacks.

When the attacker leads in a discrete time framework (Fig. 7d), the resilience metric experiences an initial sharp decline, stabilizing around -0.2 by iteration 1400. The structured intervals inherent in discrete time mitigate some of the initial damage caused by the attacker. However, the delay in defensive actions gives the attacker an early advantage, leading to moderate system degradation compared to the defender-first discrete-time scenario.

Among the four scenarios, the defender-first approach in discrete time (Fig. 7c) achieves the best resilience performance, stabilizing at -0.03 . The structured nature of discrete time allows for effective and measured defensive strategies that reduce system degradation. In contrast, the worst performance is observed in the continuous-time scenario with the defender first (Fig. 7a), where the resilience declines significantly to -4.0 . This outcome underscores the limitations of continuous-time frameworks in handling disruptions without structured decision intervals.

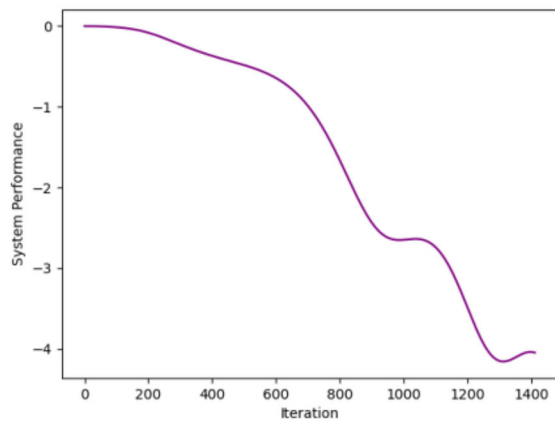
Attacker-first scenarios (Fig. 7b and d) demonstrate the critical impact of early system vulnerabilities. The resilience metric initially declines sharply in both cases due to the attacker's advantage. In continuous time, the system shows some recovery over time, whereas discrete time enables partial stabilization with structured responses. The comparison highlights the superiority of discrete-time frameworks in mitigating the impact of early attacks and maintaining overall system performance.

Defender-first strategies (Fig. 7a and c) generally outperform attacker-first strategies (Fig. 7b and d) in terms of resilience. This emphasizes the importance of proactive defenses, particularly when combined with structured decision intervals, as seen in the discrete-time framework.

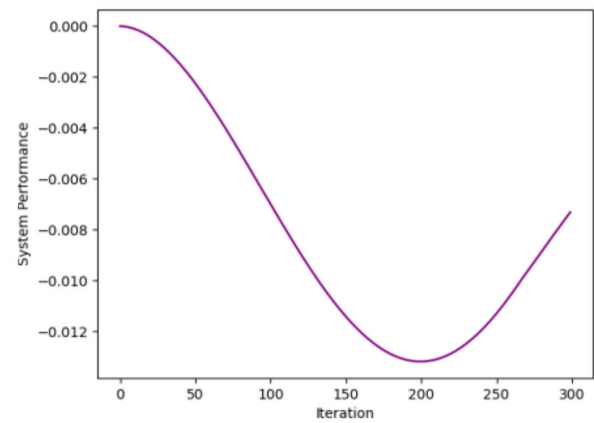
Figure 7 demonstrates the critical role of structured decision-making in enhancing resilience in a CPS power grid. Discrete-time frameworks consistently outperform continuous-time approaches by enabling proactive and measured defensive actions that minimize system degradation. The best resilience is achieved first in the discrete-time scenario of the defender, demonstrating the effectiveness of combining proactive strategies with structured decision intervals. However, continuous-time strategies, particularly in a defender-first setup, face significant limitations in mitigating disruptions and maintaining stability.

5.9 Analysis of critic losses across different scenarios

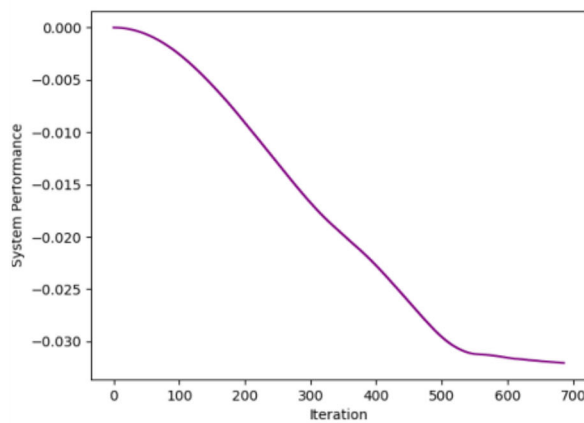
Figure 8 illustrates the evolution of Critic losses for the attacker and defender across four scenarios within a CPS



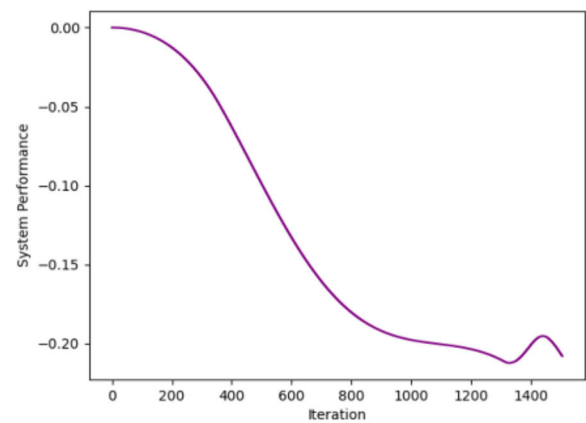
(a) System Resilience - Defender First (Continuous Time)



(b) System Resilience - Attacker First (Continuous Time)



(c) System Resilience - Defender First (Discrete Time)



(d) System Resilience - Attacker First (Discrete Time)

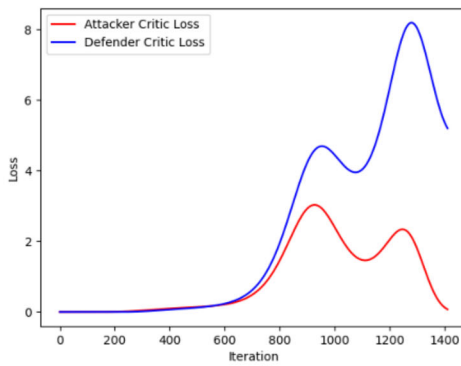
Fig. 7 Comparative analysis of resilience across different scenarios

power grid. This analysis correlates the observed critical losses with the corresponding performance metrics presented in the table, highlighting significant differences in system performance and computational overhead between continuous and discrete-time frameworks.

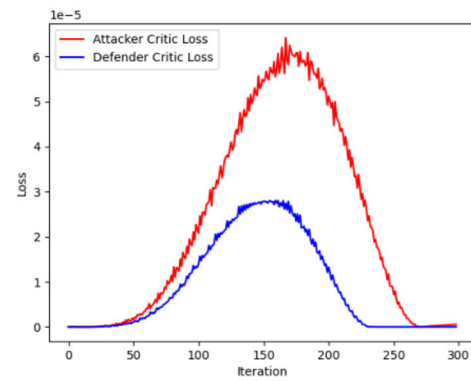
In the defender-first continuous time scenario (Fig. 8a), the defender's Critic loss rises steeply, peaking at approximately 8.1951, while the attacker's Critic loss reaches a maximum of 3.0331. The large disparity between the Critic losses reflects a substantial computational burden on the defender. This aligns with the high resilience loss (-4.0485) and system damage (4.0485) observed, indicating that the defender's strategy is computationally intensive yet less effective due to the lack of structured decision intervals in continuous time.

In the attacker-first continuous-time scenario (Fig. 8b), the attacker's Critic loss exhibits a modest peak at approximately 6×10^{-5} , while the defender's loss peaks at 4×10^{-5} . The relatively low loss values for both roles indicate reduced computational strain. However, the attacker's ability to act first and exploit system vulnerabilities results in a resilience loss of -0.0073 , reflecting minimal system disruption but also highlighting the defender's limited capacity to respond effectively in an unstructured continuous time framework.

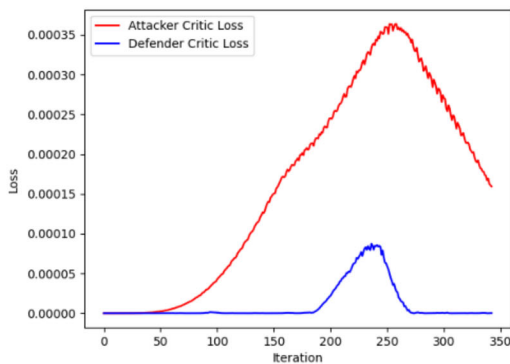
In the defender-first discrete time scenario (Fig. 8c), the defender's Critic loss peaks at approximately 0.00035, closely followed by the attacker's loss at 0.00015. These low and closely aligned values underscore the reduced computational demands and balanced dynamics facilitated by the discrete time framework. The structured decision intervals



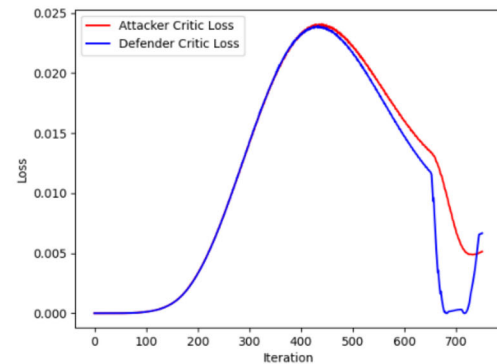
(a) Critic Losses - Defender First (Continuous Time)



(b) Critic Losses - Attacker First (Continuous Time)



(c) Critic Losses - Defender First (Discrete Time)



(d) Critic Losses - Attacker First (Discrete Time)

Fig. 8 Evolution of critic losses for attacker and defender across different scenarios

enable the defender to implement proactive measures effectively, achieving the best resilience (-0.0321) and lowest system damage (0.0321) among all scenarios.

In the attacker-first discrete time scenario (Fig. 8d), the attacker's Critic loss peaks at 0.025 , with the defender's loss reaching 0.02 . The higher Critic losses compared to the defender-first discrete scenario indicate increased computational efforts by both players. The delayed response of the defender results in a moderate loss of resilience of -0.2080 and system damage of 0.2080 . Although the discrete-time framework mitigates some computational inefficiencies, the attacker's early advantage imposes significant challenges for the defender.

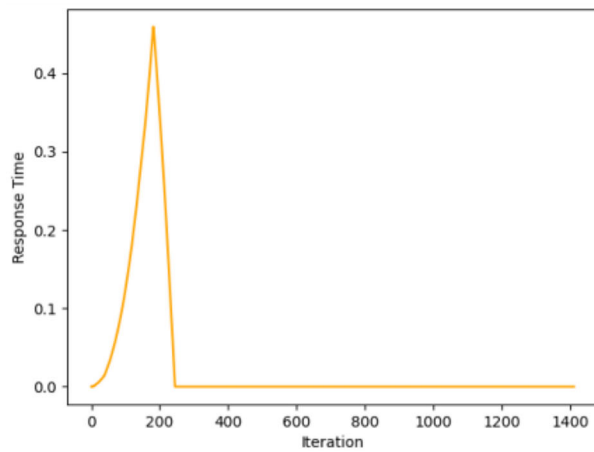
In general, Fig. 8 demonstrates the critical role of temporal frameworks and leadership roles in shaping the computational demands and system performance of CPS power grids. Discrete-time frameworks consistently outperform continuous-time approaches by reducing computational strain and enabling effective proactive defense strategies. The balanced critic losses observed in the defender-first discrete-time scenario highlight the effectiveness of combin-

ing structured decision intervals with proactive measures to achieve optimal system resilience and minimal disruption.

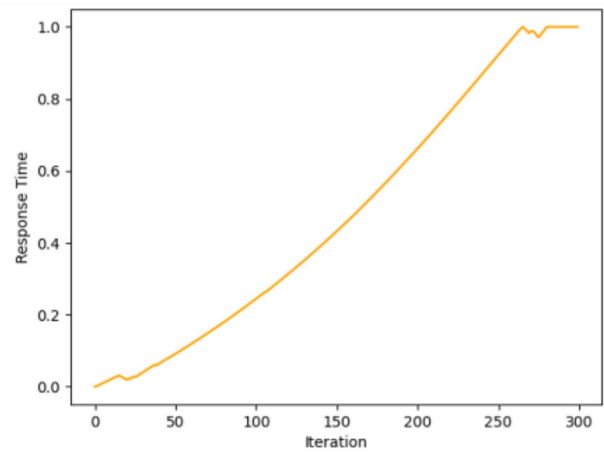
5.10 Analysis of response time across different scenarios

Figure 9 illustrates the variations in response time across four scenarios in a CPS power grid. The Figure highlights the evolution of response time under different leadership roles (defender-first vs. attacker-first) and temporal frameworks (continuous vs. discrete time). The observed trends are analyzed below in correlation with the corresponding numerical values from the response time column in the table.

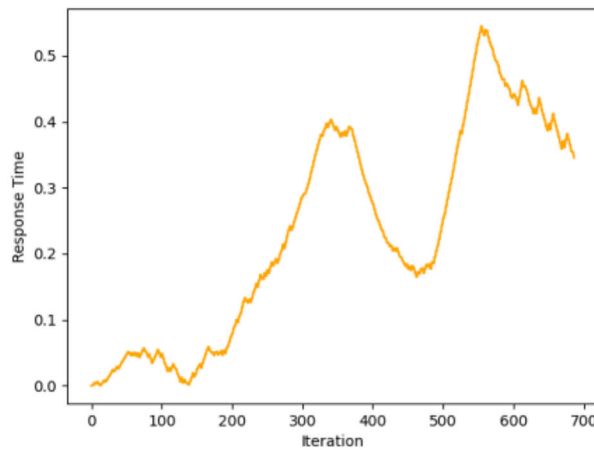
In the defender-first continuous time scenario (Fig. 9a), the response time starts at its peak value of approximately 0.45 and rapidly decreases to 0.0000 within the first 200 iterations. This immediate drop reflects the instantaneous nature of actions in continuous time frameworks. However, despite the fast response, this scenario suffers from high resilience loss (-4.0485) and significant system damage (4.0485), indicating that immediate responses do not necessarily correlate with improved system outcomes. The lack of



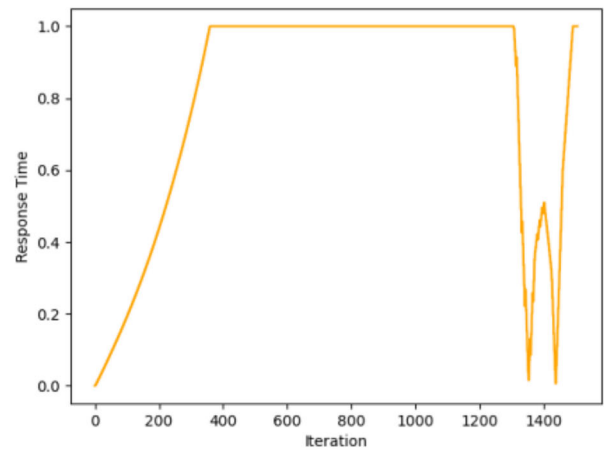
(a) Response Time - Defender First (Continuous Time)



(b) Response Time - Attacker First (Continuous Time)



(c) Response Time - Defender First (Discrete Time)



(d) Response Time - Attacker First (Discrete Time)

Fig. 9 Comparative analysis of response time across different scenarios

structured decision intervals in continuous time contributes to the ineffectiveness of the defender's strategy.

In the attacker-first continuous-time scenario (Fig. 9b), the response time exhibits a steady linear increase, reaching a maximum value of 1.0000 at the end of the iterations. This gradual escalation aligns with the attacker's ability to dictate the system's pace without immediate resistance from the defender. Despite this, the resilience loss (-0.0073) and system damage (0.0073) are minimal, highlighting that the attacker's dominant role in continuous time allows for sustained exploitation of vulnerabilities without triggering significant defensive responses.

In the defender-first discrete time scenario (Fig. 9c), the response time fluctuates, with notable peaks around iterations 200 and 500, before stabilizing at approximately 0.3455. These oscillations reflect the structured intervals of discrete time, allowing the defender to adjust their strategy iteratively. This scenario achieves the best overall performance, with the highest resilience (-0.0321) and the lowest system damage (0.0321), underscoring the advantages of proactive defense combined with discrete decision-making intervals.

In the attacker-first discrete time scenario (Fig. 9d), the response time starts at 0.0000, increases sharply to 1.0000, and fluctuates intermittently in the later stages of the iterations. This behavior demonstrates the attacker's capacity

to exploit the system early, forcing delayed and reactive responses from the defender. Consequently, the system sustains moderate resilience loss (-0.2080) and system damage (0.2080). While discrete-time frameworks reduce computational strain, the defender's delayed action diminishes the effectiveness of their response.

Continuous-time frameworks allow for immediate responses but lack the iterative refinement seen in discrete time. As a result, response times in continuous-time scenarios (Fig. 9a and b) converge to 0.0000 and 1.0000, respectively, without significant improvement in system resilience or reduction in damage. Structured decision intervals in discrete time scenarios (Fig. 9c and d) result in dynamic response time fluctuations, allowing adaptive strategies. The defender-first discrete time scenario achieves the best balance between response time (0.3455), resilience, and system damage.

Figure 9 demonstrates the critical influence of temporal frameworks and leadership roles on response time dynamics in a CPS power grid. Discrete time frameworks, particularly in the defender-first scenario, facilitate adaptive and structured responses, achieving the best system performance. In contrast, continuous-time scenarios, despite immediate actions, suffer from inefficiencies due to the absence of structured decision-making intervals. To improve system resilience and reduce damage, defensive strategies should prioritize proactive measures and minimize response delays, leveraging the iterative advantages of discrete-time frameworks.

5.11 Analysis of system damage across different scenarios

Figure 10 depicts the evolution of system damage in four scenarios in a CPS power grid. The Figure examines the impact of different leadership roles (defender-first vs. attacker-first) and temporal frameworks (continuous vs. discrete time) on system damage progression, complemented by corresponding numerical values from the table.

In the **defender-first continuous time** scenario (Fig. 10a), system damage rises steadily throughout the iterations, reaching its peak at 4.0485. This scenario demonstrates the defender's inability to prevent escalating damage despite acting first. The continuous-time framework, lacking structured decision intervals, forces the defender into reactive strategies, which fail to contain the attacker's influence effectively. The high damage levels observed are aligned with the poor resilience metric (-4.0485).

In the **attacker-first continuous time** scenario (Fig. 10b), system damage increases rapidly, peaking at 0.0073 around iteration 200 before stabilizing. The early dominance of the attacker results in a moderate level of damage, as the response of the defender is limited by the unstructured nature of continuous time. Although system damage is significantly lower

than in the defender-first scenario, the absence of proactive defense strategies highlights the vulnerabilities inherent in the framework.

In the **defender-first discrete-time** scenario (Fig. 10c), system damage rises gradually, peaking at 0.0321 toward the end of iterations. Structured decision-making intervals in discrete time allow the defender to implement effective countermeasures, minimizing system disruptions. This scenario achieves the best performance in terms of system damage and resilience (-0.0321), showcasing the advantages of discrete time in facilitating proactive defense strategies.

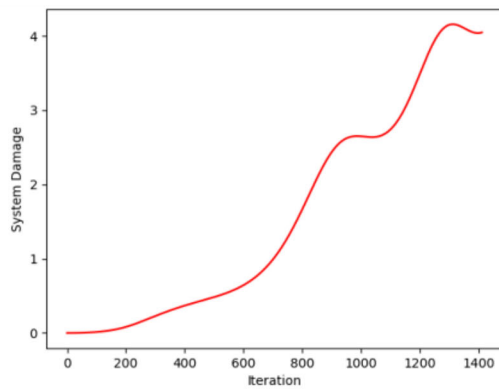
In the **attacker-first discrete-time** scenario (Fig. 10d), system damage increases steadily, reaching a peak of 0.2080. The structured nature of discrete time prevents runaway damage escalation, but the defender's delayed response to the attacker's early advantage results in moderate system damage. Although the discrete-time framework mitigates some inefficiencies, the delayed defensive action underscores the challenges of responding reactively.

The defender-first continuous-time scenario experiences the highest system damage due to the lack of structured decision intervals. Despite the defender acting first, their inability to implement proactive measures leads to significant damage escalation.

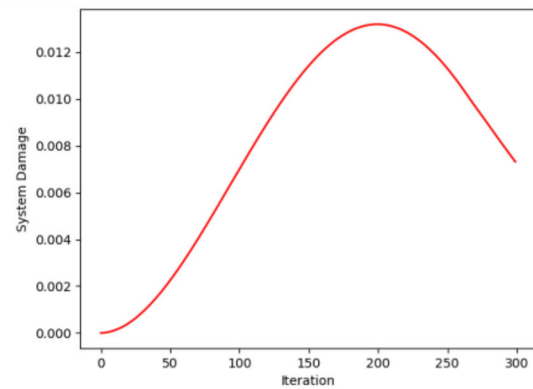
The discrete-time scenario of the defender first demonstrates the least system damage, highlighting the effectiveness of structured decision-making intervals to enable proactive defense strategies and reduce the impact of the attacker.

Attacker-first scenarios generally exhibit lower maximum system damage compared to defender-first scenarios in continuous-time frameworks. However, in discrete time, defender-first strategies consistently outperform attacker-first approaches due to their ability to leverage structured intervals for adaptive responses.

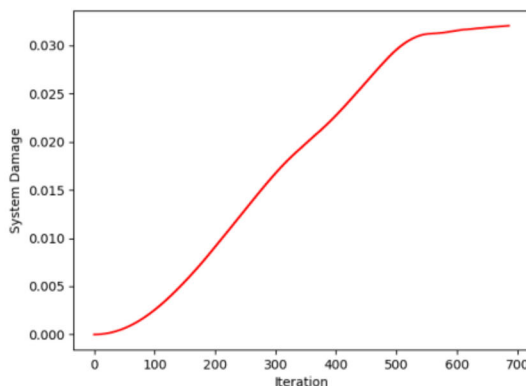
Discrete-time frameworks consistently outperform continuous-time in minimizing system damage, as structured intervals allow iterative refinement of defensive strategies. Figure 10 illustrates the critical influence of temporal frameworks and leadership roles on the progression of system damage in CPS power grids. Discrete-time frameworks, particularly in the defender-first scenario, enable structured and proactive strategies that minimize system damage and improve resilience. In contrast, continuous-time frameworks lack the iterative mechanisms required for effective defense, resulting in higher damage levels. The findings emphasize the importance of integrating structured decision-making into defensive strategies to optimize system performance under adversarial conditions.



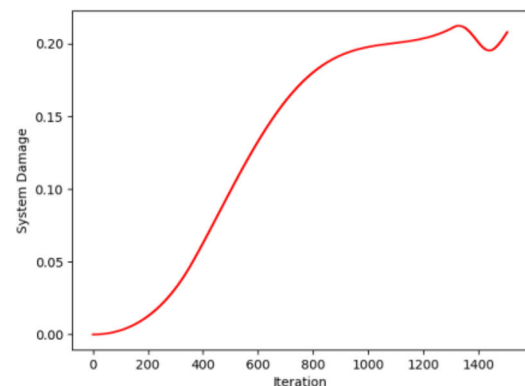
(a) System Damage - Defender First (Continuous Time)



(b) System Damage - Attacker First (Continuous Time)



(c) System Damage - Defender First (Discrete Time)



(d) System Damage - Attacker First (Discrete Time)

Fig. 10 Comparative analysis of system damage across different scenarios

5.12 Analysis of cost efficiency across different scenarios

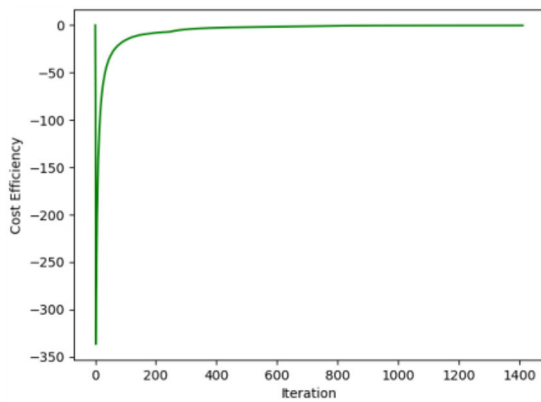
Figure 11 illustrates the evolution of cost efficiency in four scenarios in a CPS power grid. The Figure examines the influence of leadership roles (defender-first vs. attacker-first) and temporal frameworks (continuous vs. discrete time) on cost efficiency dynamics, complemented by the corresponding numerical values from the table.

In the **defender-first continuous time** scenario (Fig. 11a), cost efficiency starts at a highly negative value, rapidly increasing towards stabilization around -0.2470 . This scenario reflects the computational and resource-intensive nature of continuous-time frameworks for proactive defense. Although cost efficiency improves over time, the high loss of resilience (-4.0485) and system damage (4.0485) observed

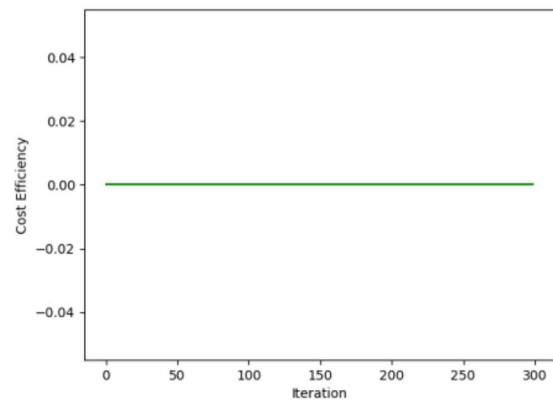
indicate that resource expenditure does not translate into effective system protection.

In the **attacker-first continuous time** scenario (Fig. 11b), cost efficiency remains nearly constant at -0.0000 throughout the iterations. This stability highlights the minimal resource utilization in this scenario. However, the attacker's preemptive actions and the defender's limited response capabilities lead to suboptimal resilience (-0.0073) and system performance, reflecting the constraints of unstructured continuous-time frameworks.

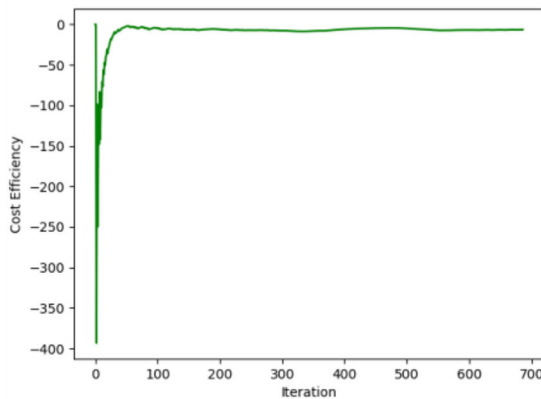
In the **defender-first discrete time** scenario (Fig. 11c), cost efficiency exhibits an initial steep increase before stabilizing around -6.4583 . The high expenditure of resources reflects the implementation of structured and proactive defense strategies enabled by discrete time intervals. Despite the cost, this scenario achieves the best system performance, with the highest resilience (-0.0321) and the lowest system



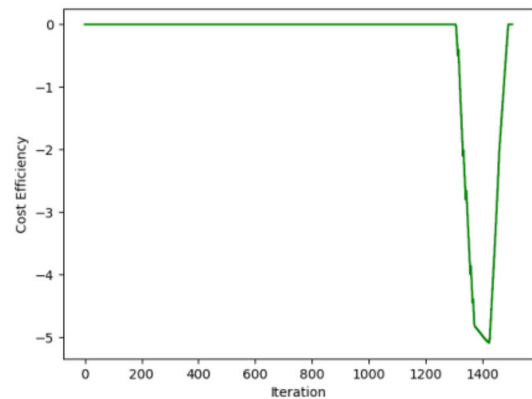
(a) Cost Efficiency - Defender First (Continuous Time)



(b) Cost Efficiency - Attacker First (Continuous Time)



(c) Cost Efficiency - Defender First (Discrete Time)



(d) Cost Efficiency - Attacker First (Discrete Time)

Fig. 11 Comparative analysis of cost efficiency across different scenarios

damage (0.0321) among all scenarios, validating the trade-off between resource allocation and system protection.

In the **attacker-first discrete time** scenario (Fig. 11d), cost efficiency fluctuates before stabilizing at -0.0000 . This stability indicates balanced resource utilization. However, the delayed defensive response results in moderate resilience loss (-0.2080) and system damage (0.2080), emphasizing the challenges of reactive strategies even in structured frameworks. Defender-first continuous-time scenarios show a rapid improvement in cost efficiency but fail to achieve meaningful system protection, highlighting the inefficiencies of unstructured frameworks.

Defender-first discrete-time scenarios demonstrate higher resource utilization but achieve optimal system performance. Structured intervals justify resource expenditure by enabling proactive and effective defense.

Attacker-first scenarios in both continuous and discrete time maintain near-zero cost efficiency, reflecting minimal resource utilization. However, the associated resilience losses and system damage highlight the limitations of reactive strategies.

The balance between resource utilization and system protection varies in all scenarios. Although structured discrete-time frameworks require higher costs, they consistently outperform continuous-time frameworks in achieving system stability and resilience.

Figure 11 highlights the critical role of cost efficiency in shaping the performance of CPS power grids. While continuous-time frameworks show rapid stabilization of cost efficiency, they fail to deliver effective system protection. In contrast, discrete-time frameworks, particularly in the defender-first scenario, demonstrate the value of structured

decision intervals, achieving optimal system performance despite higher resource expenditure. These findings underscore the importance of balancing cost efficiency with system resilience to enhance defense strategies under adversarial conditions.

5.13 Analysis of defender weights across different scenarios

Figure 12 presents the evolution of the defender weights in four scenarios in a CPS power grid. The weights of the defender represent the adaptive parameters that influence the strategic actions of the defender. This analysis highlights the impact of temporal frameworks (continuous vs. discrete time) and leadership roles (defender-first vs. attacker-first) on the evolution and stability of these weights.

In the **defender-first continuous time** scenario (Fig. 12a), the defender weights exhibit a wide range of divergence, with some weights increasing sharply and others decreasing significantly. This lack of uniformity in weight adaptation reflects the unstructured nature of continuous-time frameworks. The computational effort required for continuous updates results in high Critic losses (8.1951 for the defender) and suboptimal performance metrics, such as high system damage (4.0485) and poor resilience (-4.0485).

In the **attacker-first continuous time** scenario (Fig. 12b), the defender weights remain nearly constant, with minimal fluctuations throughout iterations. This stability indicates a limited adaptation in response to the attacker's preemptive actions. The reduced variability corresponds to the low computational effort observed in critic losses (0.0000 for the defender). However, the lack of adaptation limits the ability of the defender to mitigate attacks effectively, as reflected in the resilience loss (-0.0073).

In the **defender-first discrete time** scenario (Fig. 12c), the defender weights evolve gradually and stabilize over time. The uniform adaptation of weights reflects the structured decision intervals provided by the discrete time framework, enabling the defender to implement proactive measures effectively. This scenario achieves the best overall performance of the system, with optimal resilience (-0.0321) and minimal system damage (0.0321), validating the effectiveness of discrete time for strategic adaptation.

In the **attacker-first discrete-time** scenario (Fig. 12d), the defender weights show significant variability, with oscillations and overlapping trends throughout the iterations. These fluctuations indicate the increased computational complexity required to counter the attacker's initial advantage. Although the structured framework reduces overall inefficiencies, the delayed defensive response leads to moderate resilience loss (-0.2080) and system damage (0.2080).

In continuous-time scenarios, the absence of structured decision intervals leads to either excessive divergence

(defender-first) or limited adaptation (attacker-first) of defender weights. These dynamics result in suboptimal system performance and increased computational strain.

Discrete-time scenarios allow for gradual and structured adaptation of defender weights. The defender-first discrete time scenario, in particular, demonstrates uniform weight evolution, contributing to optimal system performance.

Defender-first scenarios typically achieve better weight adaptation compared to attacker-first scenarios. Proactive measures in defender-first scenarios promote uniform weight evolution, while attacker-first scenarios require more complex responses to counter the attacker's initial advantage.

The variability in defender weights underscores the computational challenges of different temporal frameworks. Discrete-time scenarios provide greater stability, reducing computational strain, and enabling more effective strategic adaptation.

Figure 12 underscores the critical role of defender weights in shaping the strategic outcomes of CPS power grids. Continuous-time frameworks exhibit significant limitations in adapting defender weights, resulting in poor system performance. In contrast, discrete-time frameworks, particularly in defender-first scenarios, enable structured and effective adaptation, achieving superior resilience and minimal system damage. These findings highlight the importance of structured temporal frameworks for optimizing defensive strategies in adversarial environments.

5.14 Analysis of attacker weights across different scenarios

Figure 13 illustrates the evolution of attacker weights in four scenarios in a CPS power grid. These weights represent the adaptive parameters that govern the attacker's strategy in response to the defender's actions and the system's dynamics. This analysis evaluates the influence of temporal frameworks and leadership roles on the adaptation and stability of attacker weights, correlating them with the respective performance metrics.

In the **defender-first continuous time** scenario (Fig. 13a), the attacker weights display a significant divergence, with some weights increasing sharply and others decreasing. This chaotic adaptation indicates high computational effort required to counteract the defender's preemptive actions. The steep rise and spread of attacker weights align with the observed high Critic losses (3.0331) for the attacker and the substantial system damage (4.0485) caused by the defender's reactive measures in an unstructured continuous-time framework.

In the **attacker-first continuous time** scenario (Fig. 13b), the attacker weights remain nearly constant throughout the iterations, reflecting the limited adaptation in response to the passive strategy of the attacker. This stability corresponds

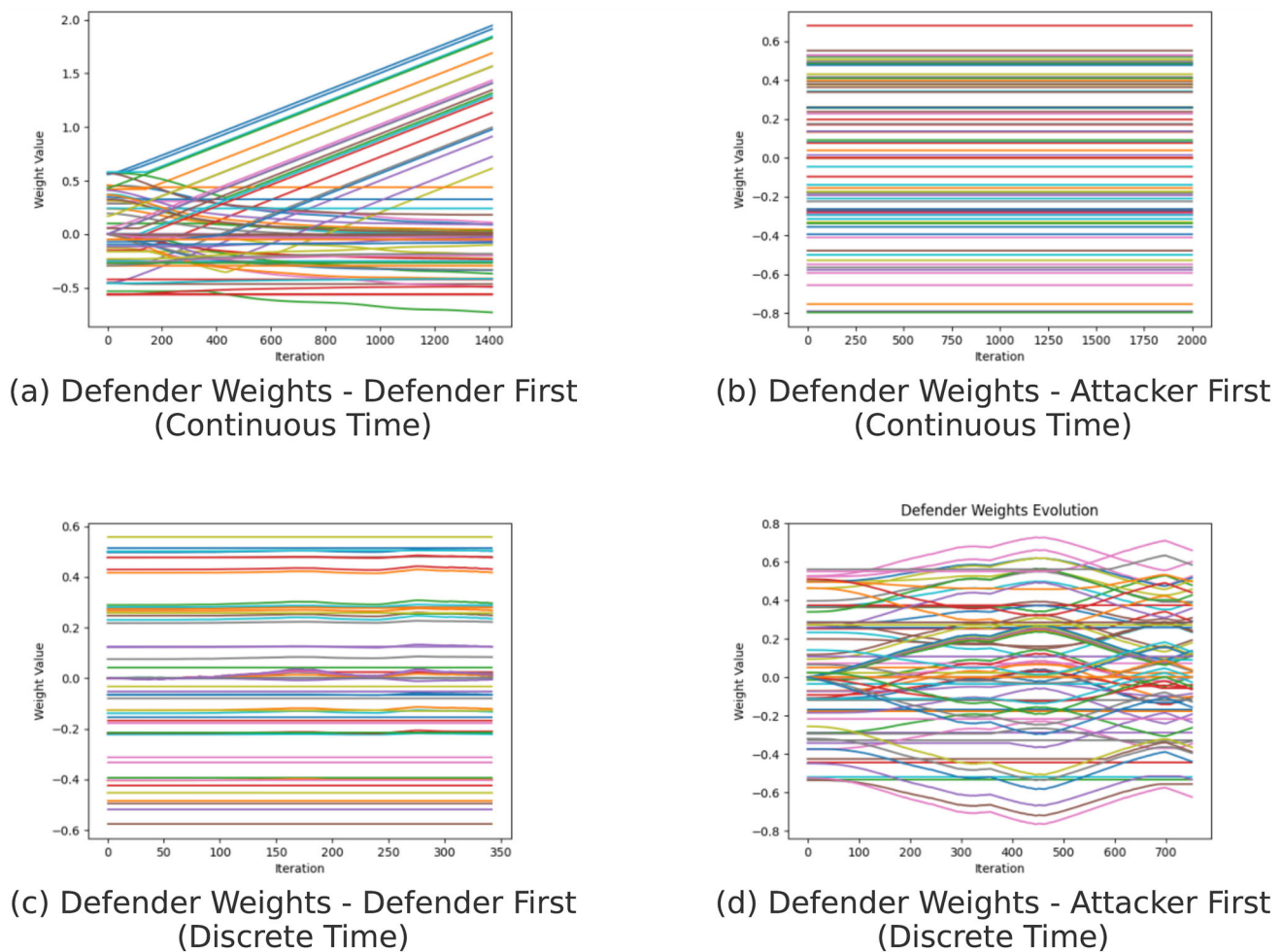


Fig. 12 Comparative analysis of defender weights across different scenarios

to minimal critical losses (0.0000) and low resilience loss (-0.0073), indicating a reduced computational effort. However, the lack of weight adaptation limits the attacker's ability to exploit vulnerabilities effectively, resulting in moderate system damage (0.0073).

In the **defender-first discrete-time** scenario (Fig. 13c), the attacker weights exhibit gradual evolution with minor fluctuations, stabilizing over time. This uniform adaptation reflects the structured decision intervals provided by the discrete-time framework, which facilitates balanced interactions between the attacker and the defender. Reduced computational effort and improved coordination lead to optimal resilience (-0.0321) and minimal system damage (0.0321).

In the **attacker-first discrete-time** scenario (Fig. 13d), the attacker weights display significant variability and oscillations throughout the iterations. This dynamic adaptation reflects the increased computational complexity required to maintain an advantage over the defender in a structured discrete-time framework. Despite the computational

challenges, this scenario achieves moderate resilience loss (-0.2080) and system damage (0.2080), highlighting the effectiveness of discrete time in managing complex interactions.

Continuous time scenarios exhibit chaotic divergence (defender-first) or limited adaptation (attacker-first) in attacker weights, reflecting the challenges of unstructured decision-making and leading to suboptimal system performance. Discrete time scenarios demonstrate more uniform and gradual adaptation of attacker weights. In particular, the defender-first discrete-time scenario achieves optimal stability and system performance, with minimal computational strain. Attacker-first scenarios require more complex weight adaptation to maintain an advantage, as evidenced by the variability in attacker weights in discrete time. In contrast, defender-first scenarios exhibit more stable weight evolution due to proactive defensive measures. The variability in the attacker weights highlights the computational demands of different frameworks. Discrete-time scenarios

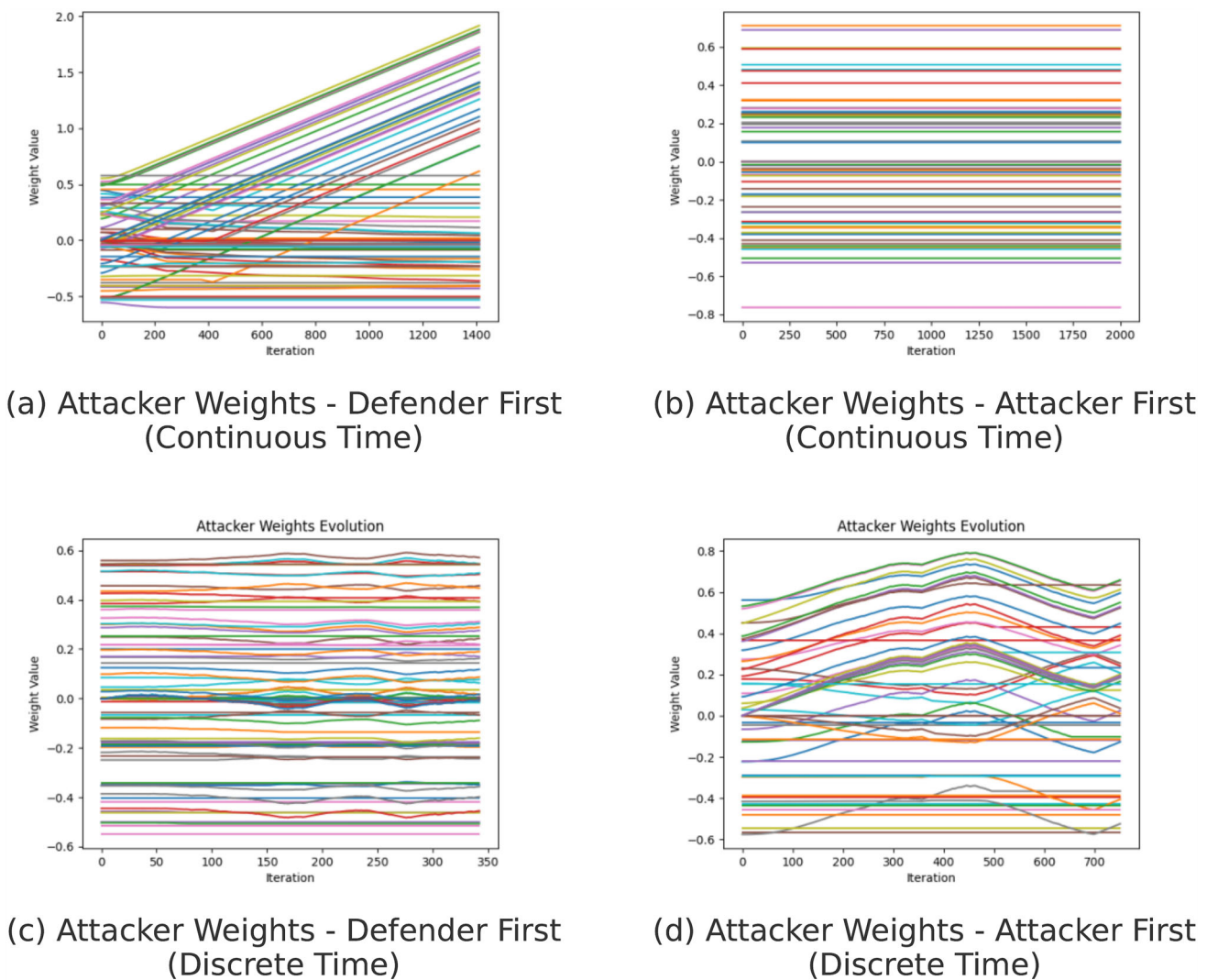


Fig. 13 Comparative analysis of attacker weights across different scenarios

provide structured adaptation, reducing computational strain, and improving strategic outcomes.

Figure 13 highlights the critical role of attacker weights in shaping the strategic dynamics of CPS power grids. Continuous-time frameworks exhibit significant limitations in the effective adaptation of attacker weights, leading to poor system performance. In contrast, discrete-time frameworks enable structured adaptation, achieving superior resilience and system stability. These findings emphasize the importance of temporal frameworks and proactive measures in optimizing adversarial strategies in CPS environments.

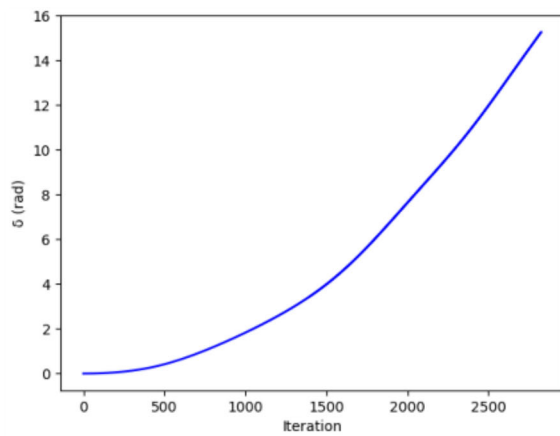
5.15 Analysis of rotor angle across different scenarios

Figure 14 depicts the evolution of the rotor angle (δ) across four scenarios in a CPS power grid. The rotor angle is a criti-

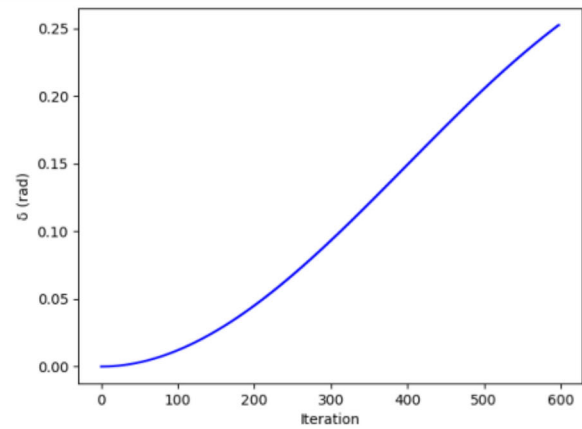
cal parameter that indicates the stability of the system, where smaller deviations reflect a more stable and resilient system. The analysis highlights the impact of temporal frameworks and leadership roles on rotor angle dynamics, correlating these with the corresponding performance metrics.

In the **defender-first continuous time** scenario (Fig. 14a), the rotor angle (δ) exhibits a rapid and unbounded increase, reaching a peak at approximately 15.2671 rad. This steep growth indicates significant instability caused by the defender's reactive measures in an unstructured framework. The lack of structured decision intervals prevents effective mitigation of the attacker's influence, resulting in high system damage (4.0485) and resilience loss (-4.0485).

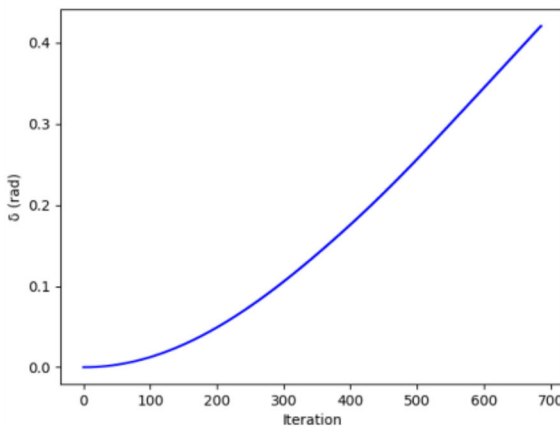
In the **attacker-first continuous time** scenario (Fig. 14b), the rotor angle stabilizes at a much smaller peak of 0.2533 rad. This indicates reduced system disruption due to the attacker's preemptive action and the defender's limited response. How-



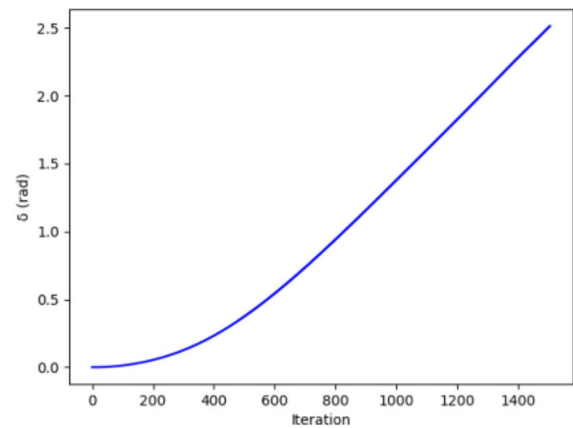
(a) Rotor Angle - Defender First (Continuous Time)



(b) Rotor Angle - Attacker First (Continuous Time)



(c) Rotor Angle - Defender First (Discrete Time)



(d) Rotor Angle - Attacker First (Discrete Time)

Fig. 14 Comparative analysis of rotor angle across different scenarios

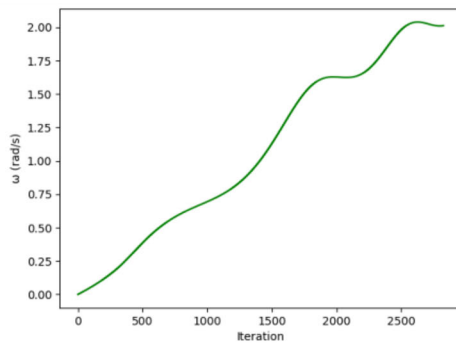
ever, while the system demonstrates improved resilience (-0.0073), the lack of defensive adaptation highlights the limitations of continuous time frameworks for proactive defense.

In the **defender-first discrete time** scenario (Fig. 14c), the rotor angle (δ) shows minimal deviation, peaking at 0.4207 rad. Structured decision intervals enable the defender to effectively counteract the attacker, maintaining system stability, and achieving the best metrics of resilience (-0.0321) and system damage (0.0321) metrics among all scenarios. The gradual and bounded increase in rotor angle reflects the advantages of proactive measures in discrete-time frameworks.

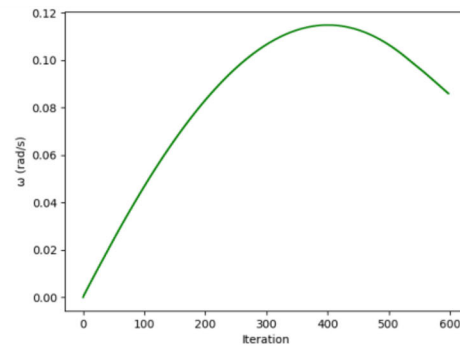
In the **attacker-first discrete-time** scenario (Fig. 14d), the rotor angle reaches a moderate peak of 2.5142 rad. The delayed response of the defender allows the attacker to exploit vulnerabilities, causing moderate system disruptions. Despite this, the structured intervals of the discrete-time framework prevent unbounded instability, resulting in a resilience loss of -0.2080 and system damage of 0.2080.

1. Continuous Time Instability: The defender-first continuous-time scenario exhibits the largest deviation of the rotor angle, highlighting the instability caused by unstructured decision-making and reactive strategies. This aligns with the high system damage and resilience loss observed.

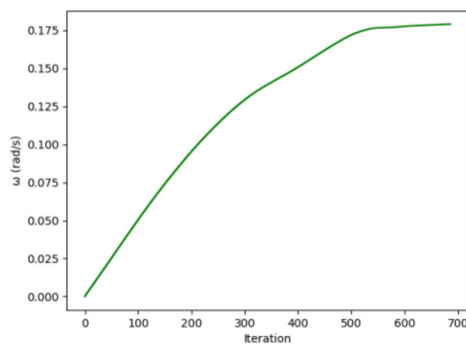
The defender-first discrete-time scenario achieves the smallest rotor angle deviation, reflecting superior stability



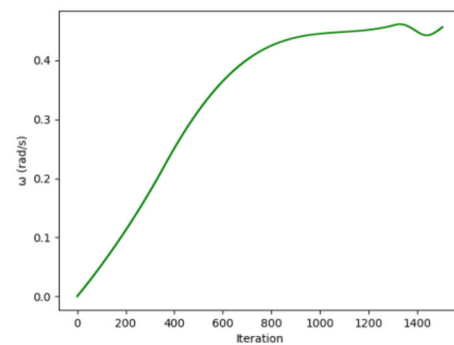
(a) Frequency Response - Defender First (Continuous Time)



(b) Frequency Response - Attacker First (Continuous Time)



(c) Frequency Response - Defender First (Discrete Time)



(d) Frequency Response - Attacker First (Discrete Time)

Fig. 15 Comparative analysis of frequency across different scenarios

and resilience. Structured decision intervals allow for proactive and effective mitigation of disruptions. Defender-first scenarios generally exhibit smaller deviations in the rotor angle compared to attacker-first scenarios. This underscores the importance of proactive defensive measures in maintaining system stability.

Continuous-time frameworks show greater susceptibility to instability, as evidenced by the larger rotor angle deviations in both scenarios. Discrete time frameworks, with their structured decision-making, consistently achieve better outcomes.

Figure 14 demonstrates the critical role of the rotor angle dynamics in assessing the stability of the system under different scenarios. Continuous-time frameworks exhibit significant limitations, with unbounded rotor angle growth reflecting poor system resilience. In contrast, discrete-time frameworks enable structured adaptation, achieving superior stability and minimal deviations from the rotor angle. These findings emphasize the importance of temporal frameworks and proactive measures in optimizing the performance of the CPS power grid.

5.16 Analysis of frequency across different scenarios

Figure 15 illustrates the evolution of frequency deviations (ω) in four scenarios in the CPS power grid. The analysis compares the results with the respective frequency values from the performance table, highlighting the impact of temporal frameworks and leadership roles.

In the **defender-first continuous time** scenario (Fig. 15a), the frequency deviation (ω) increases consistently, reaching a maximum of approximately $\omega = 2.0121$ rad/s after 2500 iterations. The steep growth in deviation reflects the unstructured nature of continuous-time frameworks, where the defender's efforts to counteract the attacker result in system instability. These results align with the high system strain and poor resilience metrics observed in this scenario.

In the **attacker-first continuous time** scenario (Fig. 15b), the frequency deviation exhibits a bell-shaped curve, peaks at $\omega = 0.12$ rad/s around iteration 400 and then stabilizes. This behavior indicates that the attacker's preemptive actions create an initial disturbance that is mitigated over time as the system reaches equilibrium. The moderate peak highlights

reduced system instability compared to defender-first strategies in continuous time.

In the **defender-first discrete-time** scenario (Fig. 15c), the frequency deviation rises to a peak of $\omega = 0.1790$ rad/s after iteration 600. The structured intervals of the discrete framework allow the defender to proactively mitigate disruptions, resulting in lower peak deviations and improved system stability. These dynamics are supported by the best resilience and system damage metrics of the scenario.

In the **attacker-first discrete time** scenario (Fig. 15d), the frequency deviation increases steadily, reaching $\omega = 0.4$ rad/s at iteration 1400. The delayed defensive response due to the attacker's first-mover advantage leads to higher peak deviations compared to the discrete scenario of the first-mover defender. This highlights the trade-offs in system stability when the attacker sets the initial conditions.

Defender-first continuous time results in the highest frequency deviation, underscoring the inefficiency of unstructured frameworks for proactive defense. The attacker-first continuous time shows reduced deviations, reflecting the system's ability to stabilize after an initial disturbance.

Both discrete-time scenarios demonstrate lower frequency deviations compared to their continuous counterparts. Structured decision intervals provide a more stable response to the system, with the defender-first scenario achieving the lowest peak deviation. Defender-first strategies generally exhibit lower frequency deviations than attacker-first strategies, highlighting the importance of proactive measures in maintaining system stability. underscores the critical role of temporal frameworks and leadership roles in shaping frequency stability within CPS power grids. Discrete time frameworks, particularly in defender-first scenarios, consistently outperform continuous time approaches by reducing frequency deviations and enhancing overall system stability.

The results of our simulations demonstrate a clear distinction between the outcomes of attacker-first and defender-first scenarios in both continuous and discrete time models. When the defender acts as the leader, preemptive defense strategies significantly influence the stability of the system. In continuous time, the defender-first scenario yields a high angular deviation of $\delta = 15.2671$ rad and a frequency deviation of $\omega = 2.0121$ rad/s, which correlates with substantial system damage (4.0485) and a cost efficiency metric of -0.2470 . These results indicate that while proactive measures by the defender mitigate immediate threats, they incur a high computational and operational cost.

In contrast, the attacker-first scenario in continuous time results in minimal system disruption, with $\delta = 0.2533$ rad and $\omega = 0.0856$ rad/s. The associated resilience metric of -0.0073 reflects the limited impact of the attacker's strategy when the defender reacts optimally. The damage to the system in this case is negligible (0.0073), indicating the effectiveness of reactive defense under these conditions.

However, the delayed response time (1.0000) highlights a critical limitation of the defender in adapting to unexpected attack patterns in real time.

In discrete-time scenarios, the outcomes differ notably due to the stepwise nature of interactions. The defender-first approach results in moderate deviations ($\delta = 0.4207$ rad and $\omega = 0.1790$ rad/s), suggesting a balanced trade-off between stability and computational efficiency. The cost efficiency improves in discrete time compared to continuous time, reflecting the benefits of iterative decision-making processes in a controlled temporal framework.

The comparative analysis underscores the importance of leadership roles in Stackelberg games for (CPS). Attacker-first scenarios demonstrate the inherent resilience of the system under optimal defensive responses, while defender-first scenarios highlight the proactive strategies needed to anticipate and neutralize threats. The findings also emphasize the role of dynamic modeling in addressing the temporal aspects of CPS defense mechanisms, validating the utility of the comparative continuous- and discrete-time modeling framework.

6 Discussion

The results of our simulations demonstrate a clear distinction between the outcomes of attacker-first and defender-first scenarios in both continuous and discrete time models. When the defender acts as the leader, preemptive defense strategies significantly influence the stability of the system. In continuous time, the defender-first scenario yields a high angular deviation of $\delta = 15.2671$ rad and a frequency deviation of $\omega = 2.0121$ rad/s, which correlates with substantial system damage (4.0485) and a cost efficiency metric of -0.2470 . These results indicate that while proactive measures by the defender mitigate immediate threats, they incur a high computational and operational cost.

To clarify the notion of "system stability" used in this study, stability is quantitatively assessed through the numerical values of the frequency deviation ω , the rotor angle deviation δ , the System Damage metric (ω^2), and the Resilience metric ($-\omega^2$). Lower magnitudes of ω and δ directly indicate improved dynamical stability, while reduced System Damage and higher (less negative) Resilience values reflect improved disturbance rejection capability.

The preemptive defense strategies referred to in the defender-first scenarios correspond to state-feedback stabilizing control policies of the form $d(t) = \pi_D(x(t))$, learned through the Actor network. In practical CPS terms, these strategies represent anticipatory damping reinforcement, corrective torque injection, or compensatory control effort applied before the attacker's action fully propagates through the system dynamics. Therefore, the defender-first

configuration does not imply a generic defensive posture, but a concrete anticipatory control policy optimized to minimize quadratic frequency deviation under worst-case attacker response.

In contrast, the attacker-first scenario in continuous time results in minimal system disruption, with $\delta = 0.2533$ rad and $\omega = 0.0856$ rad/s. The associated resilience metric of -0.0073 reflects the limited impact of the attacker's strategy when the defender reacts optimally. The damage to the system in this case is negligible (0.0073), indicating the effectiveness of reactive defense under these conditions. However, the delayed response time (1.0000) highlights a critical limitation of the defender in adapting to unexpected attack patterns in real time.

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Importantly, the study offers a comprehensive framework by integrating both discrete and continuous time models, which overcomes a major limitation observed in previous works such as Feng et al. [12] and Li et al. [22], where only one temporal framework was typically considered. This unified modeling allows for more flexible and realistic evaluations of cyber-defense strategies across different operational timescales.

Additionally, our implementation of Stackelberg game theory in combination with DRL provides a novel and practical contribution to the CPS security field. Previous studies (e.g., [24, 26]) often addressed such game-theoretic models in theoretical or simplified settings, whereas our approach is validated in complex, nonlinear CPS environments. This not only enhances scientific understanding but also proves its real-world applicability.

This contribution is achieved by embedding the Stackelberg leader–follower structure directly into the learning architecture. Instead of solving the Stackelberg equilibrium analytically, we approximate the leader's optimal commitment and the follower's best-response mapping through cou-

pled Actor–Critic networks. The leader's policy is optimized while anticipating the adaptive response of the follower, whose strategy is simultaneously learned through temporal-difference value updates.

Consequently, the theoretical Stackelberg equilibrium is not treated as a static analytical concept, but is operationalized as a data-driven, dynamically learned equilibrium emerging from repeated state-feedback interactions under nonlinear CPS dynamics. This mechanism enables real-time adaptation, scalability, and applicability to complex cyber–physical environments where closed-form solutions are infeasible.

Another key insight from our results is the superior performance of defender-first strategies in discrete-time scenarios, which achieve optimal balance between proactiveness and computational feasibility. Our findings suggest a measurable benefit of anticipatory defense strategies compared to reactive ones in CPS security, contributing to an area that remains underexplored in existing literature.

Furthermore, our DRL-based architecture introduces adaptability that static methods lack. It allows for continuous self-improvement of defense policies in response to evolving threats, addressing the call in the literature for more flexible and real-time security systems (cf. [4, 18]).

Finally, our experimental setup bridges the gap between theory and practice. Through extensive Python simulations, we quantify improvements in resilience, reductions in system damage, and computational cost efficiency—key metrics that enhance scientific reproducibility and operational viability. Compared to earlier theoretical or limited-scope studies (e.g., [16, 41]). Our approach contributes a working model that could support and inspire further development and benchmarking efforts in this domain.

While the present study focuses on a two-player Stackelberg game involving one defender and one attacker, the proposed framework can be naturally extended to scenarios with a single defender and multiple heterogeneous attackers. In such a setting, each attacker acts as an independent follower that adapts its strategy based on the observed system state evolution, while the defender commits to a single policy anticipating either the aggregate or worst-case attacker response. This formulation would allow the modeling of simultaneous, uncoordinated, or heterogeneous attack behaviors in large-scale CPS. Investigating multi-follower Stackelberg games combined with multi-agent reinforcement learning constitutes an important direction for future work.

7 Conclusion and future work

This study presents a novel integration of Stackelberg games and DRL in the design of dynamic defense strategies for (CPS), particularly in the context of power grid security. By

incorporating both continuous and discrete time models into a unified framework, our approach addresses a critical gap in the prior literature, where most studies (e.g., [12, 22]) treat these temporal paradigms in isolation.

Our results demonstrate that the role of leadership in Stackelberg interactions—whether attacker-first or defender-first—substantially impacts system resilience and cost efficiency. Notably, the defender-first strategy in discrete time produced the lowest system damage and highest resilience, validating the superiority of anticipatory defense strategies over purely reactive ones. This insight builds upon but also significantly extends, previous studies that did not explore role reversals or leadership dynamics in depth (e.g., [24]; Mishra et al. [26]).

Furthermore, the proposed hybrid framework exhibits real-time adaptability, overcoming the static limitations of prior defense strategies. By leveraging DRL, the system learns and evolves dynamically in response to new attack patterns—fulfilling a need repeatedly emphasized in the literature for adaptive, multi-layered security architectures (e.g., [4]).

Crucially, we validate our framework through extensive Python-based simulations, yielding quantifiable improvements in resilience, reduced damage, and lower costs. Unlike earlier studies focused primarily on theoretical models, our results reflect practical applicability, thus bridging the gap between theory and implementation.

Although our proposed framework demonstrates promising results, certain limitations must be acknowledged. First, the model considers a single attacker and defender, limiting the scope of multi-agent dynamics commonly observed in real-world CPS environments. Second, we did not incorporate the physical constraints of power systems, such as voltage thresholds or power flow limits, which may influence the practical applicability of the results. Additionally, the costs and delays associated with anticipatory defenses, as observed in continuous-time scenarios, highlight potential challenges in real-time deployment. These constraints offer valuable directions for future investigation.

Building on the foundations laid in this study, numerous research opportunities arise that could significantly extend the impact and applicability of our framework within the CPS security domain.

First, expanding the model to include multiple attackers and defenders would allow for the exploration of collaborative, coordinated, and adversarial behaviors across complex networked environments. This level of scalability remains largely unexplored in the literature and could reveal emergent strategic dynamics vital to the protection of large-scale infrastructures. Another important extension concerns the modeling of uncertainty in the attacker's payoff structure. In many realistic CPS environments, the defender does not possess complete information about the attacker's type,

objectives, or cost parameters. This motivates the adoption of a Bayesian Stackelberg game formulation, where the defender (leader) optimizes its strategy with respect to a belief distribution over possible attacker types.

In such a setting, each attacker type would be characterized by type-dependent reward functions and cost sensitivities, while the defender maintains and updates a belief distribution over these types based on observed actions. The resulting equilibrium concept would correspond to a Bayesian Stackelberg equilibrium, where the leader commits to a strategy that maximizes expected utility under type uncertainty.

Integrating this formulation into the proposed DRL-based architecture would require augmenting the state representation with belief variables and incorporating type-conditioned policy learning. This extension would allow the framework to capture incomplete-information scenarios and heterogeneous adversarial behaviors, further enhancing its realism and applicability in large-scale CPS environments.

Second, introducing adversaries that utilize both discrete and continuous-time attack strategies could test the system's resilience under mixed threat regimes, a scenario much closer to real-world complexities. This would challenge existing defense policies and promote the development of more robust, time-aware response mechanisms.

Third, incorporating operational constraints—such as voltage thresholds, power flow limits, and load fluctuations—would significantly improve the fidelity of simulations, aligning them with real-world power system conditions. Such enhancements would bridge the current gap between theoretical control models and deployable technologies.

Furthermore, combining DRL with supervised and unsupervised learning approaches opens avenues for more sample-efficient learning and improved decision-making. This integration could lead to faster convergence, greater policy generalization, and a better ability to adapt in rapidly evolving threat landscapes.

In addition, there is a growing need for long-term, dynamic resilience indicators that go beyond short-term costs or damage. Designing novel performance metrics tailored to persistent and adaptive adversaries would offer a richer understanding of system robustness and enable more rigorous benchmarking of future defense frameworks.

Finally, applying this framework to other CPS domains—such as intelligent transportation systems, industrial automation, and smart healthcare—would validate its generality and versatility. These sectors each present unique cyber-physical challenges, and tailored adaptations of our approach could contribute to sector-specific security solutions.

These directions collectively aim to enhance the scalability, realism, and resilience of CPS defense strategies. By addressing these research challenges, future studies can build on our work to deliver high-impact, practical solutions against emerging and sophisticated cyber-physical threats.

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Author Contributions Anastasios Raptis contributed to the design of the game-theoretic framework, the problem formulation, and the reinforcement learning methodology. He also developed the simulation code and conducted the experiments. Stefanos Gritzalis and Athanasios Yannacopoulos jointly supervised the research. Stefanos Gritzalis contributed to the strategic formulation and review of the cybersecurity aspects, while Athanasios Yannacopoulos provided mathematical validation and reviewed the theoretical analysis.

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Data Availability The data that support the findings of this study are included in the manuscript. The simulation code used in this work is available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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